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0607/23

May/June 2018

45 minutes

Additional Materials: Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.

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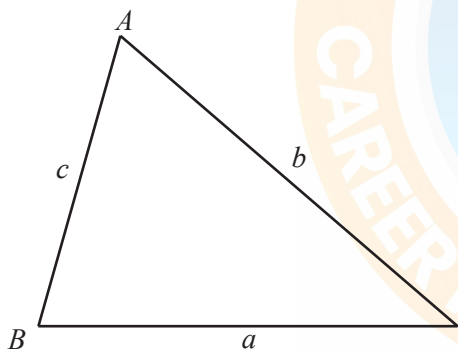
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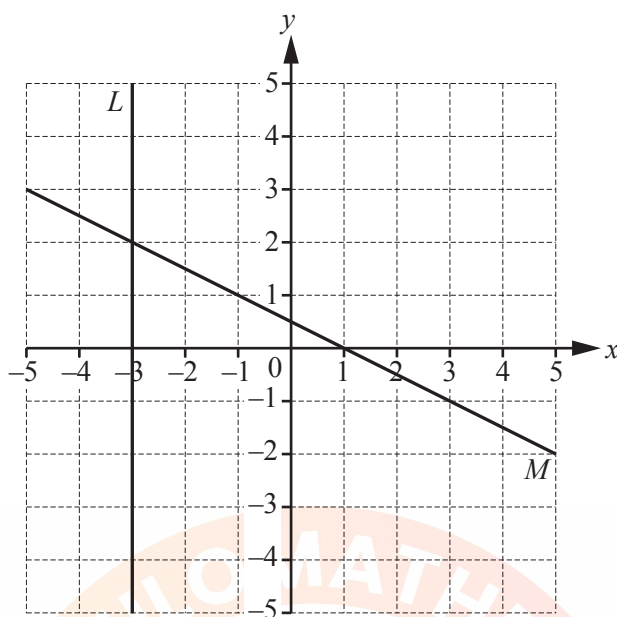
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1



(a) Write down the equation of line L .

..... [1]

(b) Write down the co-ordinates of the point of intersection of line L and line M .

(.....,) [1]

(c) Find the gradient of line M .

..... [2]

2 Find the highest common factor (HCF) of 96 and 60.

..... [1]

3 Expand and simplify $5(2x + 3y) - 3(4y - 2x)$.

..... [2]

- 4 Write down the value of 17^0 .

..... [1]

5 $v = \frac{uf}{u-f}$

Find v when $u = 30$ and $f = 10$.

$v =$ [2]

- 6 (a) Find a fraction, n , that satisfies this inequality.

$$\frac{5}{7} < n < \frac{6}{7}$$

$n =$ [1]

- (b) Write down an irrational number, m , that satisfies this inequality.

$$4 < m < 7$$

$m =$ [1]

- 7 Q is the point $(3, 7)$ and $\vec{PQ} = \begin{pmatrix} -6 \\ 3 \end{pmatrix}$.

- (a) Find the co-ordinates of P .

(.....,) [2]

- (b) Find $|\vec{PQ}|$.
Give your answer in its simplest surd form.

..... [3]

- 8 Work out $(5.6 \times 10^{-7}) - (7.8 \times 10^{-8})$.
Give your answer in standard form.

..... [2]

- 9 Kim has a piece of rope 18 metres long.
He cuts the rope into two pieces.
The lengths of the pieces are in the ratio 1 : 5.

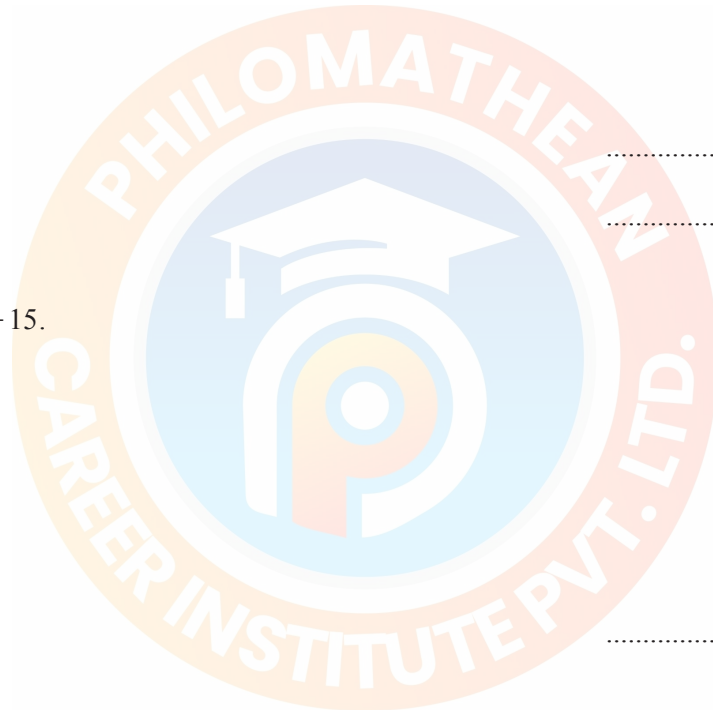
Calculate the length of each piece.

..... m

..... m [2]

- 10 Solve $3 - x \geq 2x + 15$.

..... [2]



- 11 Jamil has a biased 6-sided die.
He rolls it 350 times.
The results are shown in the table.

Number on die	1	2	3	4	5	6
Frequency	20	50	72	68	56	84

- (a) Find the relative frequency of getting a 2 with Jamil's die.

..... [1]

- (b) Explain why your answer to **part (a)** is a good estimate of the probability of getting a 2.

.....[1]

- (c) Estimate the number of times Jamil will get a 2 if he rolls the die 1400 times.

..... [1]

- 12 (a) On the grid, sketch the graph of $y = \sin x^\circ$ for $0 \leq x \leq 360$.



[2]

- (b) The point $(a, 0.5)$ is on the graph of $y = \sin x^\circ$.

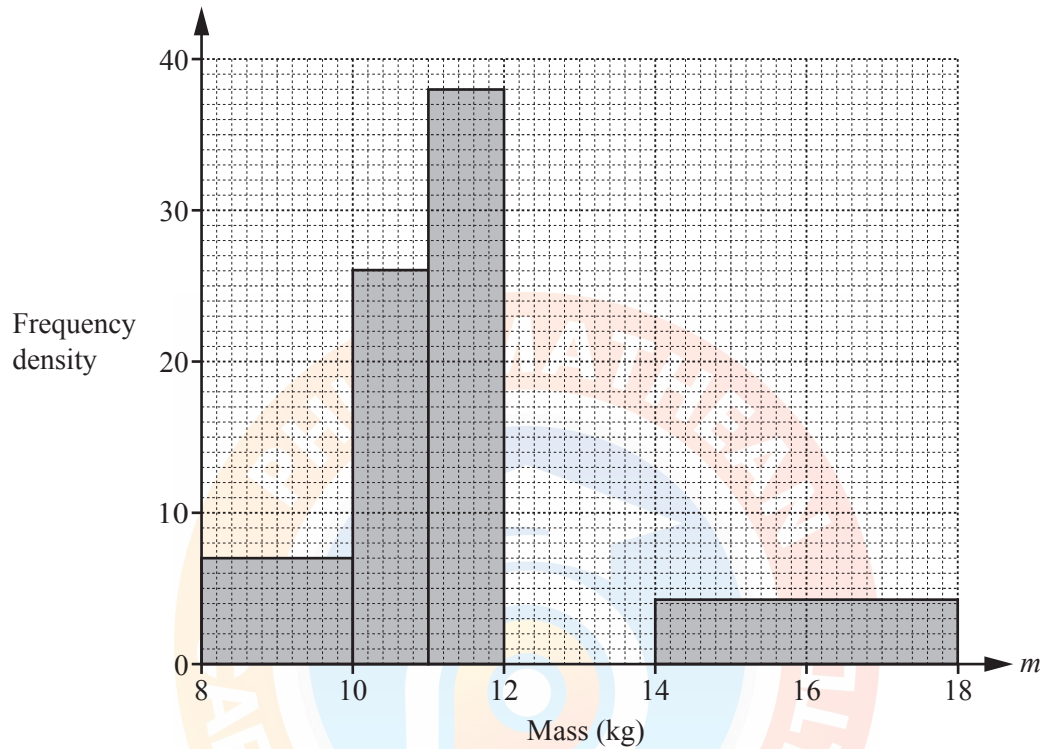
Find the two possible values of a .

$a = \dots\dots\dots$ or $a = \dots\dots\dots$ [2]

- 13 The masses, m kg, of some watermelons are measured.
The results are shown in the table.

Mass (m kg)	$8 < m \leq 10$	$10 < m \leq 11$	$11 < m \leq 12$	$12 < m \leq 14$	$14 < m \leq 18$
Frequency	p	26	38	24	17

Part of the histogram to show this information is shown below.



- (a) Complete the histogram. [2]
(b) Find the value of p .

$$p = \dots\dots\dots [1]$$

Questions 14 and 15 are printed on the next page.

- 14 Rearrange this formula to make x the subject.

$$y = \frac{ax}{bx+c}$$

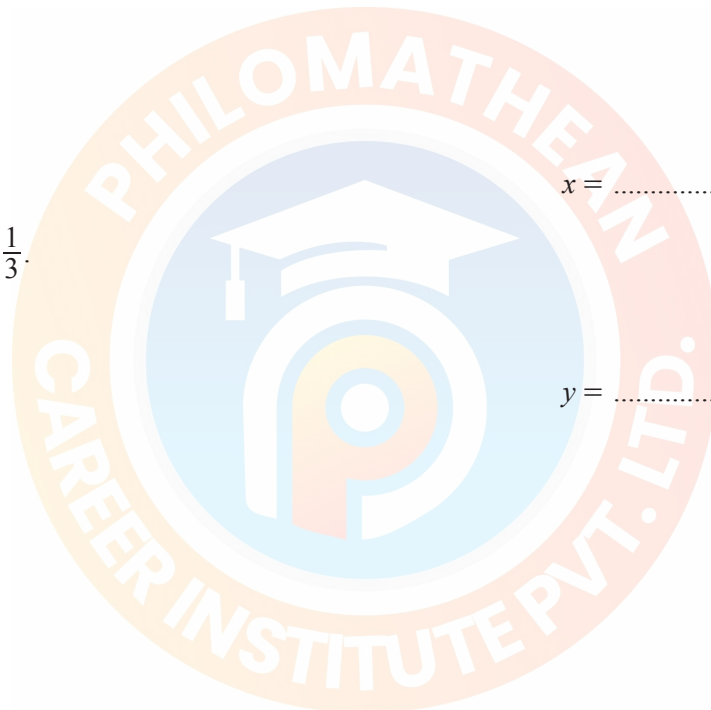
$$x = \dots\dots\dots [3]$$

- 15 (a) Solve $3 \log 2 - 2 \log 5 = \log x$.

$$x = \dots\dots\dots [3]$$

- (b) Solve $\log_y 4 = \frac{1}{3}$.

$$y = \dots\dots\dots [1]$$



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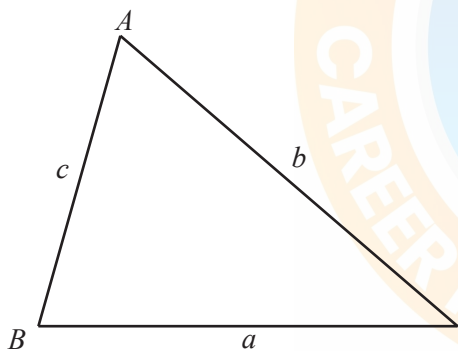
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$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1 Work out.

(a) $(0.3)^2$

..... [1]

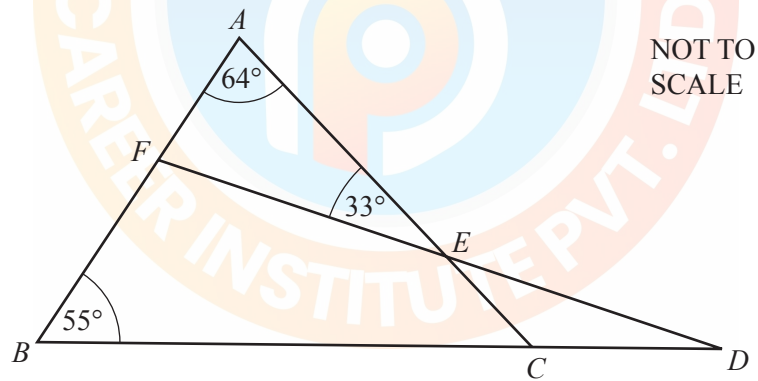
(b) $\frac{4}{9} - \frac{1}{6}$

..... [2]

2 Divide 360 in the ratio 7 : 2.

....., [2]

3



ABC is a triangle.

FED and BCD are straight lines.

Work out angle EDC .

Angle EDC = [2]

- 4 Expand and simplify.

$$4(3x + y) - 3(x - 2y)$$

..... [2]

- 5 Sacha drove 425 km from home at an average speed of 100 km/h.

- (a) Calculate the time for the journey giving your answer in hours and minutes.

..... h min [2]

- (b) The return journey took 3 hours and 55 minutes.
She started at 21 56.

At what time did she arrive home?

..... [2]

- 6 (a) Write down the integer solutions to this inequality.

$$-2 \leq 2x < 8$$

..... [2]

- (b) Solve $2 + 2x > 5x + 14$.

..... [2]

- 7 Work out $(5.2 \times 10^{18}) - (2.4 \times 10^{17})$.
Give your answer in standard form.

..... [2]

- 8 A map is drawn to a scale of 1 cm to 5 km.

- (a) On the map, the distance between two towns is 4.8 cm.

Find the actual distance between the towns.

..... km [1]

- (b) An island has an area of 75 km^2 .

Find the area of the island on the map.

..... cm^2 [2]

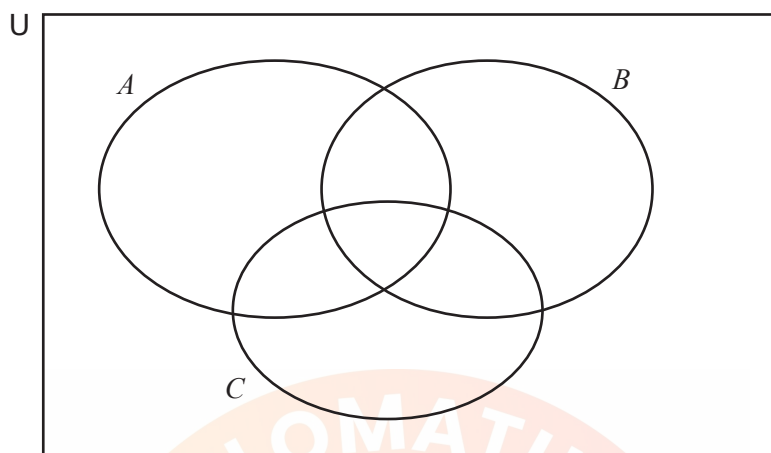
- 9 Factorise completely.

$$2x^2 - 18$$

..... [2]

- 10 $U = \{\text{integers from 1 to 12}\}$
 $A = \{1, 2, 4, 5, 12\}$
 $B = \{2, 3, 4, 6, 10\}$
 $C = \{1, 2, 8, 9, 10\}$

(a) Complete the Venn Diagram.



[2]

(b) Find $n(A \cap (B \cup C)')$.

[1]

- 11 The point A has co-ordinates $(3, 8)$.
The point B has co-ordinates $(7, 0)$.

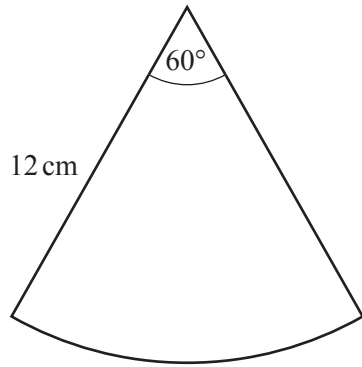
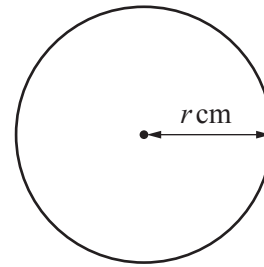
(a) Find the co-ordinates of the midpoint of AB .

(.....,) [1]

(b) Find the equation of the perpendicular bisector of AB .
Write your answer in the form $y = mx + c$.

$y = \dots\dots\dots$ [3]

12

NOT TO
SCALE

The sector and the circle have the same area.

The angle of the sector is 60° .

The radius of the sector is 12 cm and the radius of the circle is r cm.

Work out the value of r .

Give your answer as a surd in its simplest form.

$r =$ [3]

13 Rearrange this formula to make b the subject.

$$A = \frac{(a+b)}{2}h$$

$b =$ [3]

Question 14 is printed on the next page.

14 (a) Find the value of $\log_{25} 5$.

..... [1]

(b) Simplify $\log 63 - 2 \log 3$.

..... [2]



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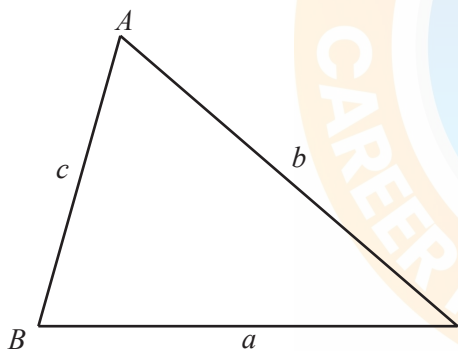
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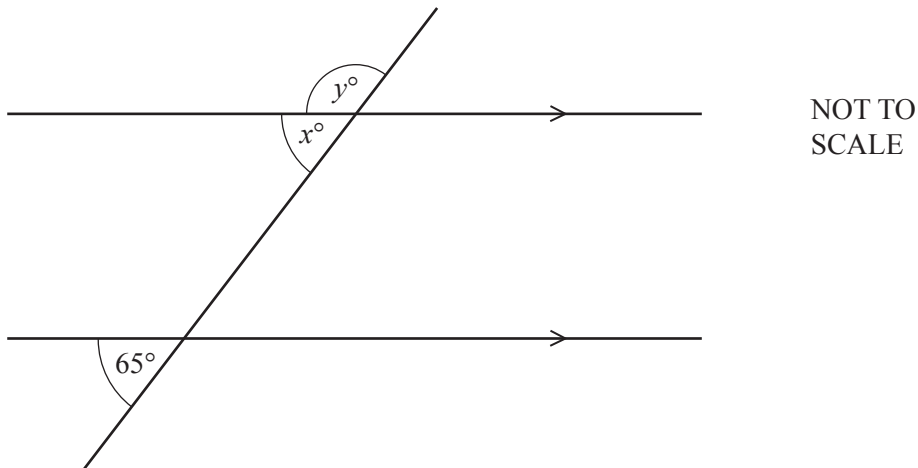
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1



Find the value of x and the value of y .

$x =$
 $y =$ [2]

2 A regular polygon has 40 sides.

Find the size of one exterior angle.

..... [2]

3 A is the point $(1, 5)$ and B is the point $(6, 2)$.

Find the column vector $\overrightarrow{AB}$.

$\begin{pmatrix} \\ \end{pmatrix}$ [2]

4 $t = 3p^2$

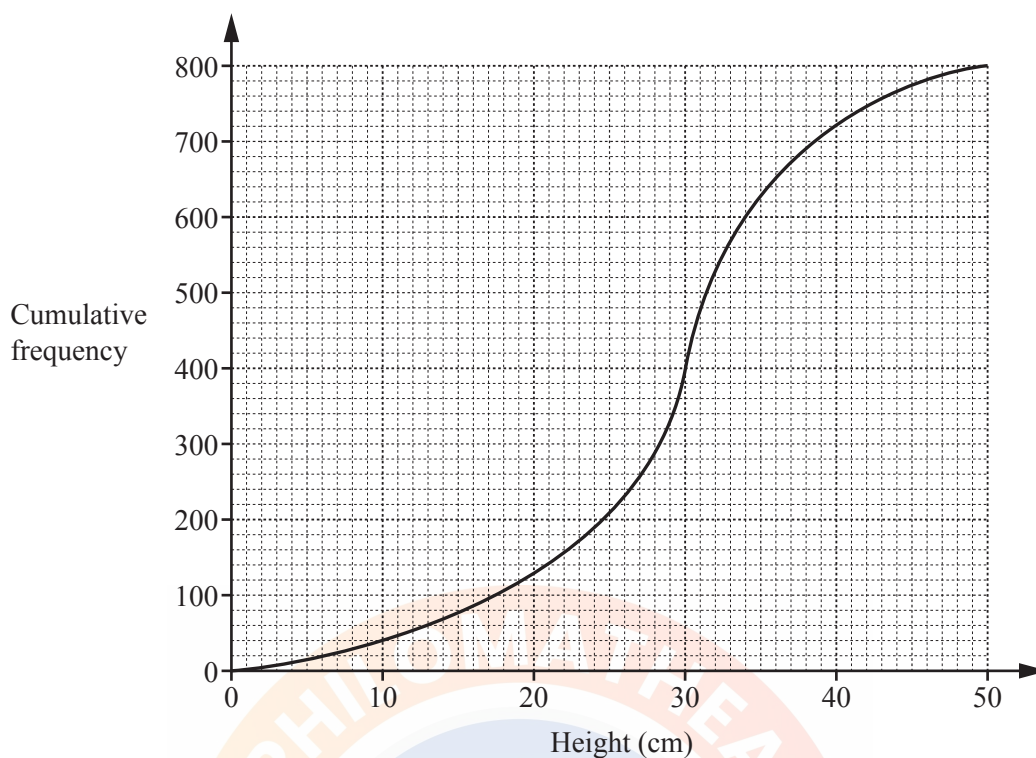
(a) Find the value of t when $p = 4$.

$t =$ [1]

(b) Re-arrange the formula to write p in terms of t .

$p =$ [2]

5



The cumulative frequency curve shows some information about the heights of 800 plants.

Find

(a) the median,

..... cm [1]

(b) the upper quartile.

..... cm [1]

6 Work out $\frac{4}{5} \div 1\frac{1}{2}$.

..... [2]

7 A car travels 85 km in 50 minutes.

Find the average speed of the car, giving your answer in km/h.

..... km/h [2]

- 8 Solve the simultaneous equations.

$$a + b = 16$$

$$2a - b = 17$$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots [2]$$

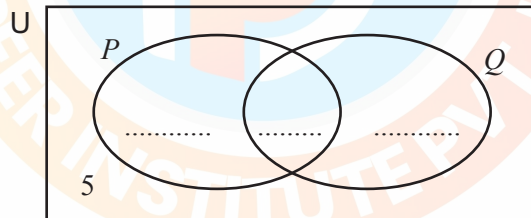
- 9 Find the equation of the line parallel to the line $y = 3 - x$ that passes through the point $(0, 7)$.

$$\dots\dots\dots [2]$$

- 10 Work out the value of $\left(\frac{1}{27}\right)^{-\frac{1}{3}}$.

$$\dots\dots\dots [1]$$

11



$$n(U) = 25 \quad n(P) = 10 \quad n(Q) = 17 \quad n(P \cup Q)' = 5$$

Complete the Venn diagram.

[2]

- 12 Factorise completely.

$$ab - a - b + 1$$

$$\dots\dots\dots [2]$$

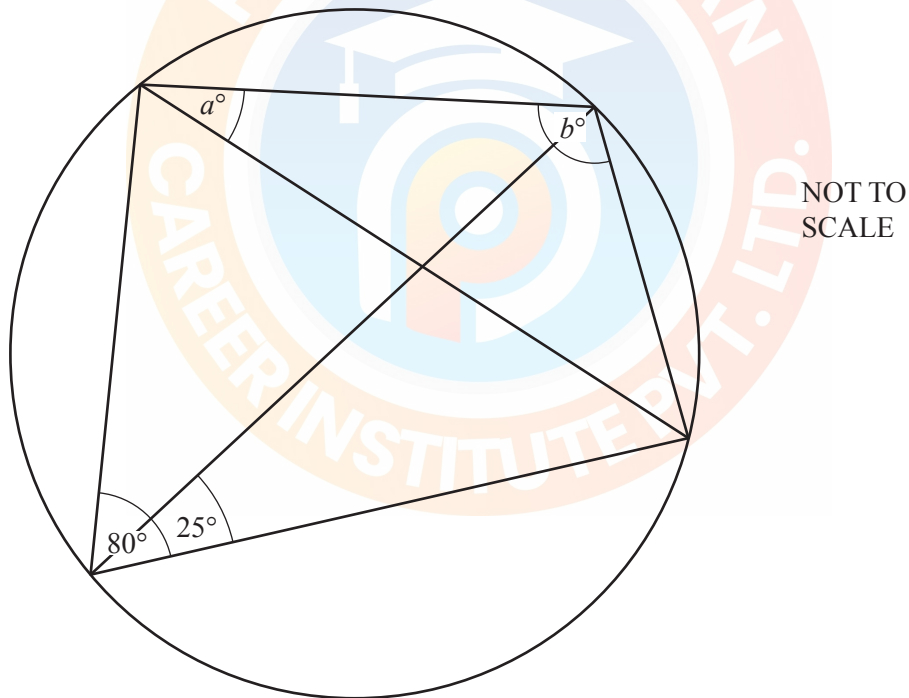
- 13 Work out $1.1 \times 10^{30} + 1.1 \times 10^{29}$, giving your answer in standard form.

..... [2]

- 14 Find the highest common factor (HCF) of $8p^4q^8$ and $4p^3q^{10}$.

..... [2]

15



The diagram shows a cyclic quadrilateral.

Find the value of a and the value of b .

$a =$

$b =$ [2]

- 16 Rationalise the denominator.

$$\frac{1}{\sqrt{5}-1}$$

..... [2]

- 17 y is inversely proportional to $\sqrt{x+4}$.
When $x = 5$, $y = 12$.

Find y in terms of x .

$y =$ [2]

- 18 Simplify. $\frac{y^2-9}{xy+3x}$

..... [3]

Question 19 is printed on the next page.

19 (a) $2 \log x = 3 \log 4$

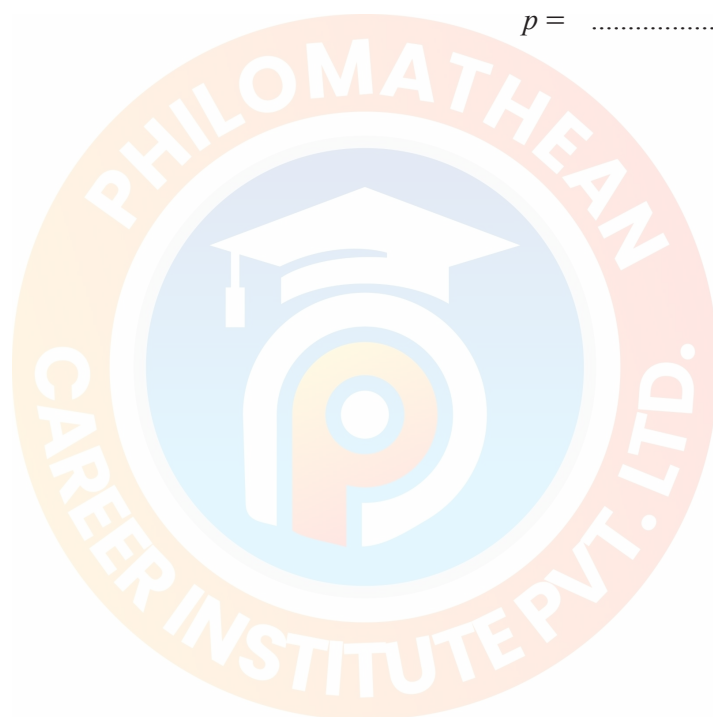
Find the value of x .

$x = \dots\dots\dots$ [2]

(b) $\log x + \log u - \log v = \log p$

Find p in terms of x , u and v .

$p = \dots\dots\dots$ [1]



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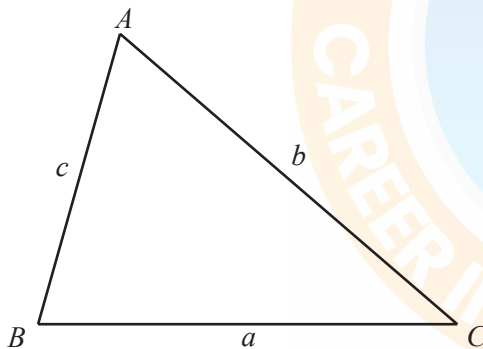
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- 1 Work out.

$$(-2)^3$$

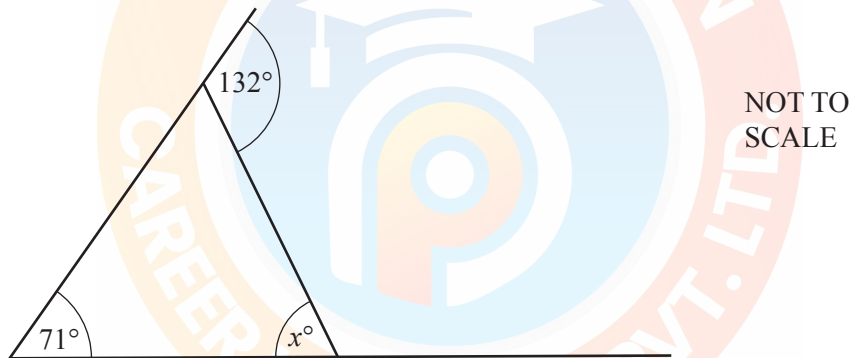
..... [1]

- 2 $f(x) = 1 - 3x$

Find the value of $f(-1)$.

..... [1]

- 3



Find the value of x .

$x =$ [2]

- 4 Expand the brackets and simplify.

$$2(3x - 1) + 3(1 - 2x)$$

..... [2]

5 A quadrilateral has

- two pairs of parallel sides
- all sides the same length
- no right angles.

Write down the mathematical name of this quadrilateral.

..... [1]

6

16 10 11 15 10 12 14 13 17 10 15

Find the median of these eleven numbers.

..... [1]

7 Work out.

$$5\frac{2}{5} \times 1\frac{2}{3}$$

..... [3]

- 8 Work out the following.
Give each answer in standard form.

(a) $(1 \times 10^1) + (2 \times 10^{-2})$

..... [2]

(b) $(1 \times 10^1) \div (2 \times 10^{-2})$

..... [2]

- 9 A bag contains 2 blue balls, 3 red balls and 5 green balls only.
John takes a ball out of the bag at random.
He records the colour and puts the ball back in the bag.
Flavia takes a ball out of the bag at random and records the colour.

Find the probability that both balls are red.

..... [2]

10 $\mathbf{a} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 2 \\ -8 \end{pmatrix}$

- (a) Find $\mathbf{a} - 3\mathbf{b}$.

$\begin{pmatrix} \\ \end{pmatrix}$ [2]

- (b) Work out $|\mathbf{a}|$.

..... [2]

- 11 A travel agent has the following exchange rates.

$$£1 = \$1.25$$

$$£1 = €1.20$$

- (a) Change £200 into dollars (\$).

\$ [1]

- (b) Change \$100 into euros (€).

€ [2]

- 12 The point A has co-ordinates $(1, 3)$ and the point B has co-ordinates $(4, 1)$.
 B is the midpoint of the line AC .

Find the co-ordinates of the point C .

(.....,) [2]

- 13 Make a the subject of $s = ut + \frac{1}{2}at^2$.

$a =$ [3]

- 14 Factorise completely.

$$6ac - 9bc - 8ad + 12bd$$

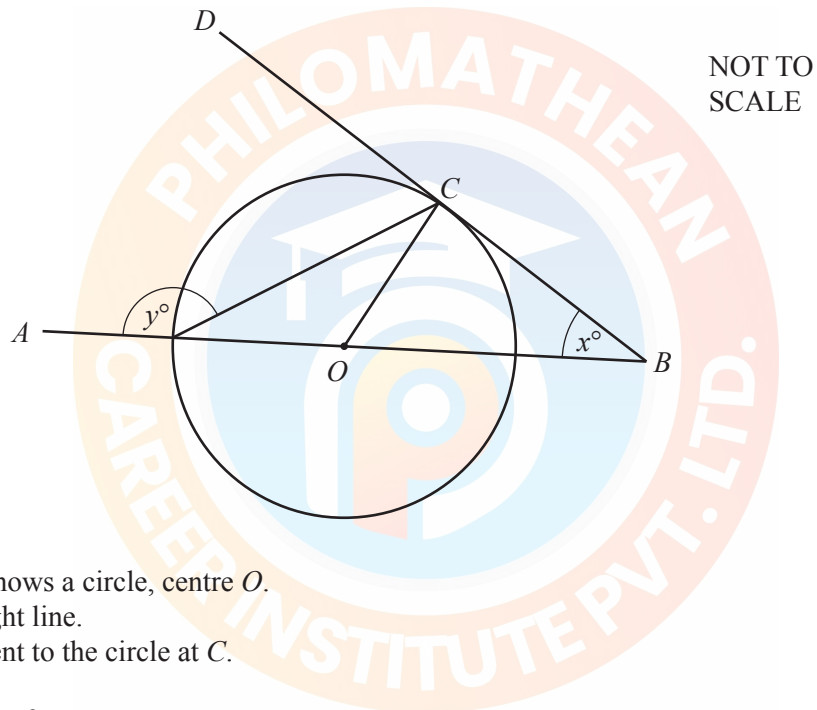
..... [2]

- 15 Erica walks 13 km in 2 hours.
She then runs at a speed of 12 km/h for 45 minutes.

Find her average speed in km/h for the whole journey.

..... km/h [3]

16



The diagram shows a circle, centre O .
 AOB is a straight line.
 BCD is a tangent to the circle at C .

Find y in terms of x .

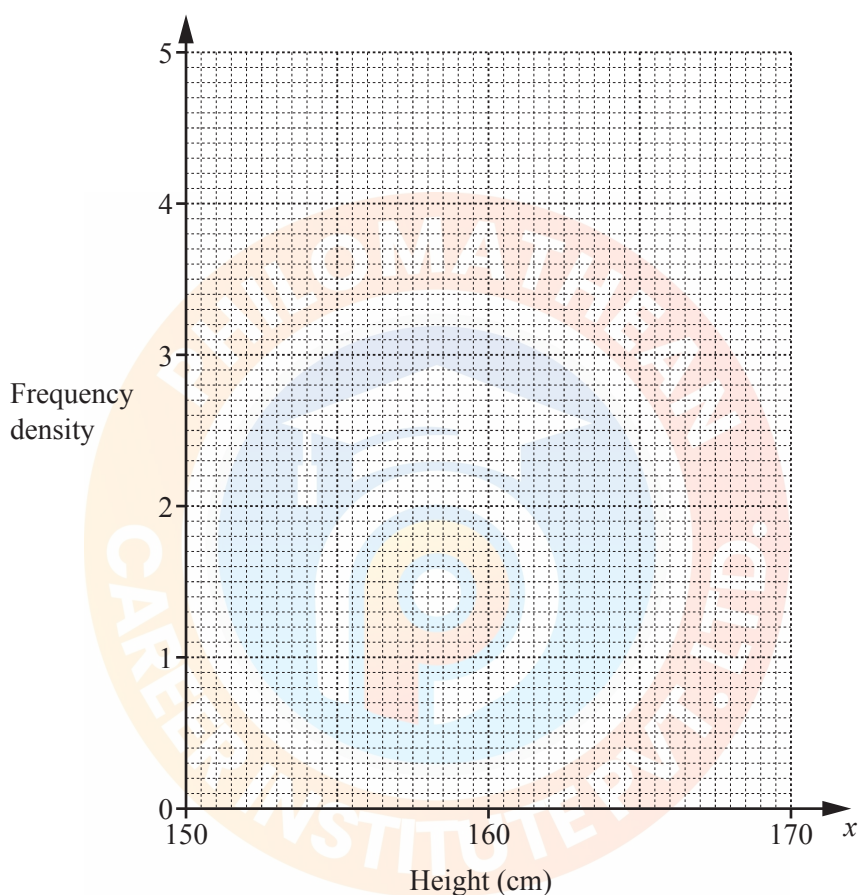
$y =$ [3]

Question 17 is printed on the next page.

17 The table shows the heights, x cm, of some students at a school.

Height (x cm)	Frequency
$150 < x \leq 160$	8
$160 < x \leq 165$	20
$165 < x \leq 170$	24

On the grid below, draw a histogram to show this information.



[3]

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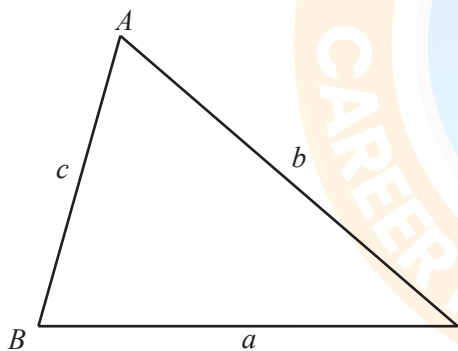
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Volume, V , of cylinder of radius r , height h . $V = \pi r^2 h$

Volume, V , of cone of radius r , height h . $V = \frac{1}{3}\pi r^2 h$

Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 A cuboid has a square base of side 10 cm and a volume of 1200 cm^3 .

Work out the height of the cuboid.

..... cm [2]

2 $\mathbf{p} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

- (a) Find $\mathbf{p} + \mathbf{q}$.

$\begin{pmatrix} \\ \end{pmatrix}$ [1]

- (b) A is the point (2, 7).

The point A is translated to the point B by the vector $\mathbf{p} + \mathbf{q}$.

Find the coordinates of B .

(.....,) [2]

- 3 Work out $\frac{3}{4} \div 2\frac{1}{2}$.

Give your answer as a fraction in its lowest terms.

..... [3]

- 4 A truck of length 10 m passes a gate of length 2 m.
The speed of the truck is 8 m/s.

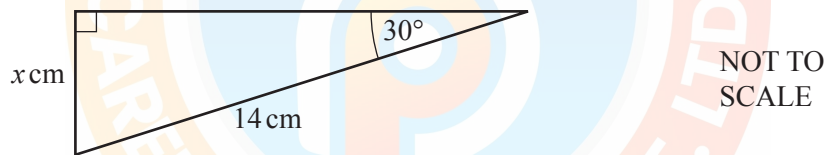
Find the time the truck takes to completely pass the gate.

..... s [2]

- 5 Find the volume of a cone with radius 3 cm and perpendicular height 8 cm.
Give your answer in terms of π .

..... cm^3 [2]

6



Work out the value of x .

$x =$ [3]

- 7 Simplify.

(a) $\frac{15w^{15}}{3w^3}$

..... [2]

(b) $(125y^6)^{\frac{2}{3}}$

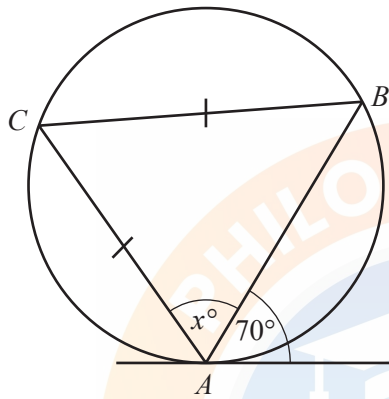
..... [2]

8 $A = 2\pi rh + 3\pi r^2$

Rearrange the formula to write h in terms of π , r and A .

$h = \dots\dots\dots$ [2]

9



NOT TO
SCALE

A , B and C are points on a circle.
 TA is a tangent to the circle at A .
 $CA = CB$ and angle $BAT = 70^\circ$.

Work out the value of x .

$x = \dots\dots\dots$ [2]

10 When Jack sells a computer for \$264 he makes a profit of 20%.

Work out the price Jack paid for the computer.

\$ $\dots\dots\dots$ [2]

- 11 y is inversely proportional to $\sqrt{x}$.
When $x = 9$, $y = 2$.

Find y in terms of x .

$$y = \dots\dots\dots [2]$$

- 12 $3 \log y = 2 \log x - \log w$

Find y in terms of x and w .

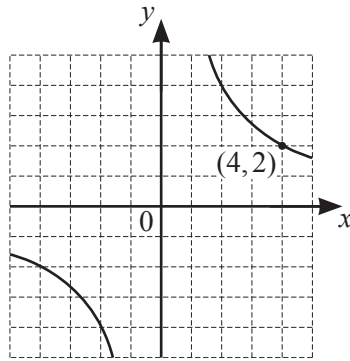
$$y = \dots\dots\dots [3]$$

- 13 Rationalise the denominator.

$$\frac{9}{\sqrt{7}-2}$$

$$\dots\dots\dots [2]$$

14



In the diagram, the graph passes through the point (4, 2).

Write down the equation of the graph.

[2]

15 Simplify.

$$\frac{3-a}{3p-6t-ap+2at}$$

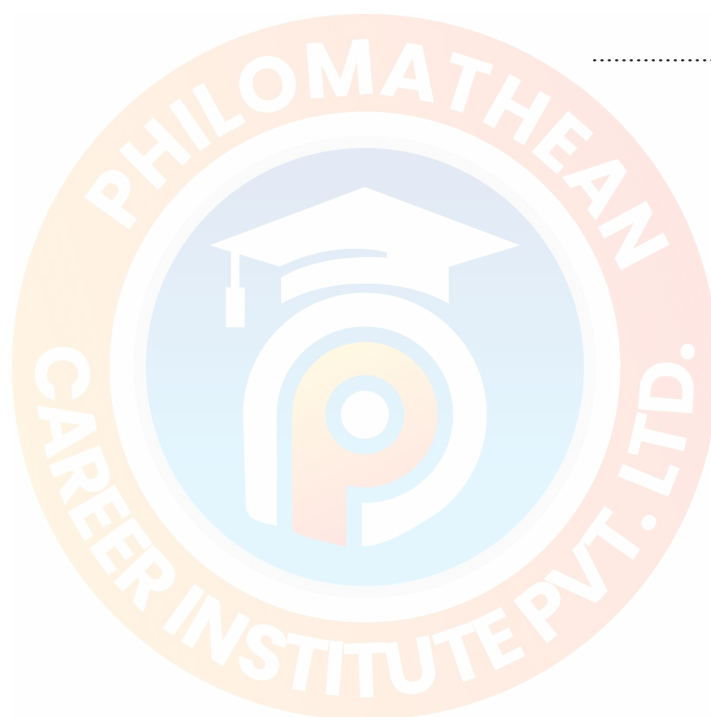
[3]

Question 16 is printed on the next page.

16 Write as a single fraction in its simplest form.

$$\frac{1}{x-3} - \frac{2}{x}$$

..... [3]



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Curved surface area, A , of cone of radius r , sloping edge l . $A = \pi rl$

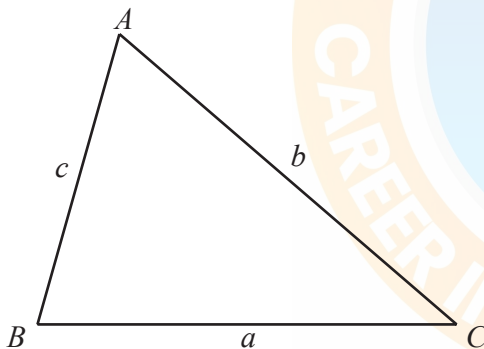
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 31 37 39 49 51 53 77 87

From this list write down **all** the prime numbers.

..... [2]

- 2 Work out 15% of 600.

..... [2]

- 3 Work out.

(a) 0.06×0.12

..... [1]

(b) 0.2^3

..... [1]

(c) $\frac{0.4}{0.08}$

..... [1]

- 4 A bag contains red balls, blue balls and green balls only.
There are twice as many blue balls as green balls.
There are twice as many red balls as blue balls.
There are 16 blue balls in the bag.

Find the total number of balls in the bag.

..... [2]

- 5 Dippi buys 5 burgers and 4 bags of chips for a total cost of \$8.10 .
Burgers cost \$1.10 each.

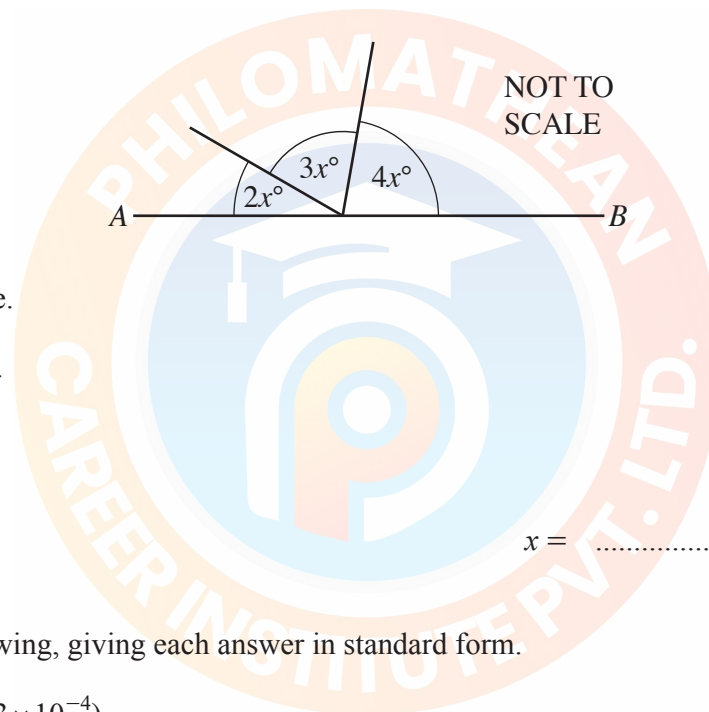
Find the cost of one bag of chips.

\$ [3]

6

AB is a straight line.

Find the value of x .



$x =$ [2]

- 7 Work out the following, giving each answer in standard form.

(a) $(4.3 \times 10^4) \times (3 \times 10^{-4})$

..... [2]

(b) $(6 \times 10^{-2}) + (3 \times 10^{-3})$

..... [2]

- 8 Solve the simultaneous equations.

$$\begin{aligned} 3x + 2y &= -1 \\ 7x - y &= 26 \end{aligned}$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots [3]$$

- 9 The interior angle of a regular polygon is 150° .

Find the number of sides of this polygon.

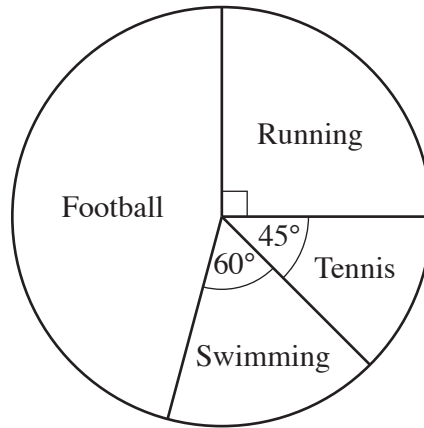
$$\dots\dots\dots [3]$$

- 10 Rearrange the formula to make x the subject.

$$y(x+4) = 2$$

$$x = \dots\dots\dots [2]$$

11



The pie chart shows the favourite sports of all the students at a school.
180 students chose running as their favourite sport.

Work out

(a) the total number of students at the school,

[1]

(b) the number of students that chose football as their favourite sport.

[2]

12 Factorise.

$$2x^2 - 3x - 5$$

[2]

13 Solve.

$$(x-4)(x+3) > 0$$

..... [2]

14 A is the point $(1, 7)$ and B is the point $(4, 13)$.

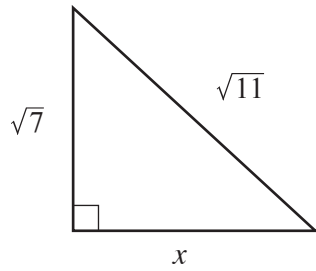
Find the equation of the perpendicular bisector of AB in the form $y = mx + c$.



$y =$ [5]

Question 15 is printed on the next page.

15

NOT TO
SCALEFind the value of x . $x =$ [2]

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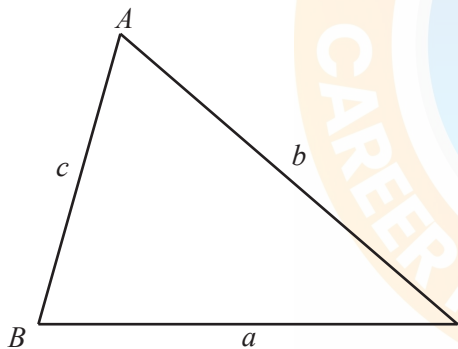
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 (a) Write 0.047 996 correct to 4 decimal places.

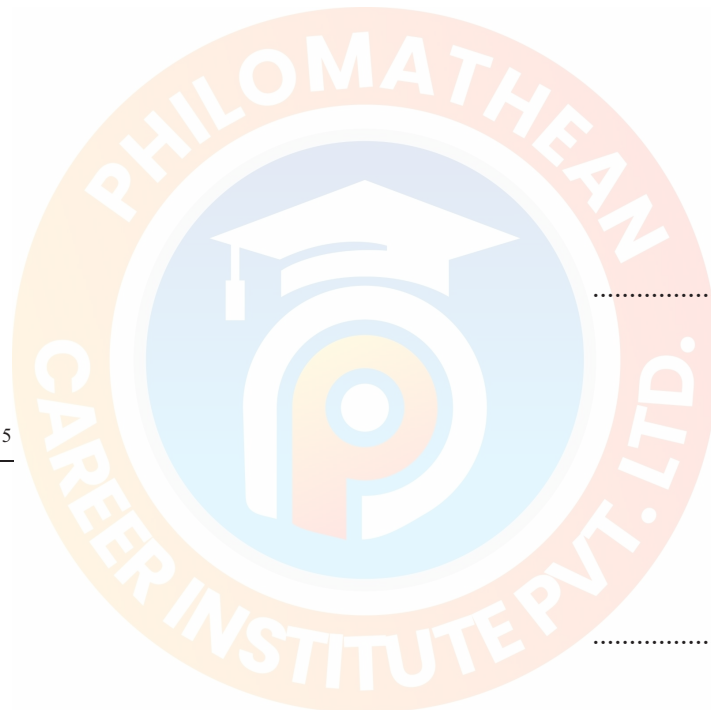
..... [1]

- (b) Write 60 449 correct to 3 significant figures.

..... [1]

- 2 Work out $4\frac{1}{4} - 1\frac{5}{6}$.

Give your answer as a mixed number in its simplest form.



- 3 Simplify.

$$\frac{a^2 \times a^5}{a^3}$$

..... [3]

..... [2]

- 4 (a) Write down the mathematical name of the quadrilateral that has rotational symmetry of order 2 but no lines of symmetry.

..... [1]

- (b) Write down the mathematical name of the quadrilateral that has exactly one line of symmetry.

..... [1]

5 Solve.

$$9 - 2x \leq 5(x + 6)$$

..... [3]

6 A biased four-sided spinner is spun 150 times.

The number of times that the spinner lands on each number is shown in the table.

Number on spinner	1	2	3	4
Frequency	34	63	27	26

(a) Write down the relative frequency of the spinner landing on 2.

..... [1]

(b) Explain why it is reasonable to use your answer to **part (a)** as the probability of this spinner landing on 2.

..... [1]

(c) The spinner is spun 3000 times.

Find the expected number of times that the spinner lands on 2.

..... [2]

7 Divide 96 cm in the ratio 5 : 3.

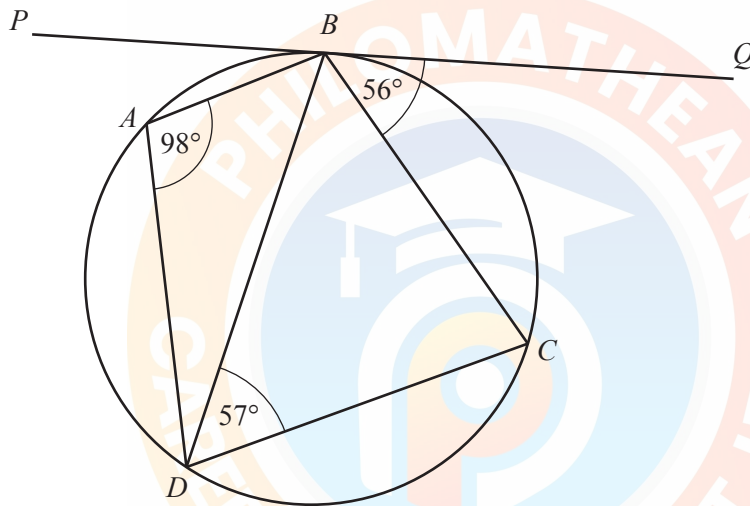
..... cm , cm [2]

- 8 A is the point $(-2, 4)$ and B is the point $(7, 1)$.

Find the length of AB giving your answer in its simplest surd form.

..... [4]

9



NOT TO
SCALE

A , B , C and D are points on the circle.
 PBQ is a straight line.

- (a) Find angle DCB , giving a reason for your answer.

Angle DCB = because

..... [2]

- (b) Is PBQ a tangent to the circle?
Give a reason for your answer.

..... because

..... [1]

10 Solve the simultaneous equations.

$$2x + 3y = 5$$

$$y = 3x + 9$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots [3]$$

11 The table shows some trigonometric ratios, each correct to 3 decimal places.

	Sine	Cosine	Tangent
40°	0.643	0.766	0.839
70°	0.940	0.342	2.747

Use this information to find

(a) $\sin 110^\circ$,

..... [1]

(b) $\tan 320^\circ$.

..... [1]

12 Factorise completely.

(a) $4x^2y - 6xy^2$

..... [2]

(b) $9x^2 - 1$

..... [1]

13 Solve.

(a) $\log_x 9 = 2$

$x =$ [1]

(b) $2 \log x - \log 4 = \log 9$

$x =$ [2]

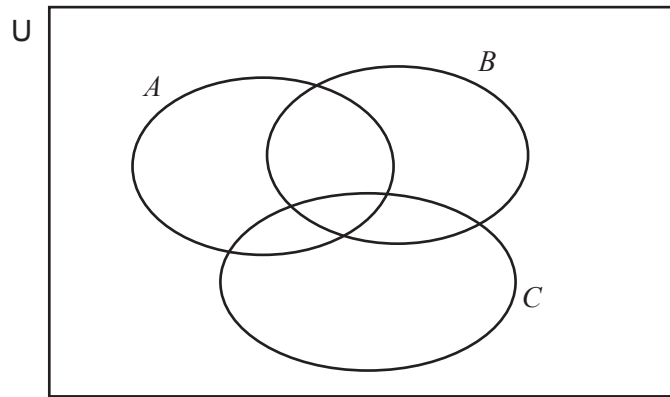
14 y varies inversely as the square root of x .
When $x = 25$, $y = 6$.

Find y in terms of x .

$y =$ [2]

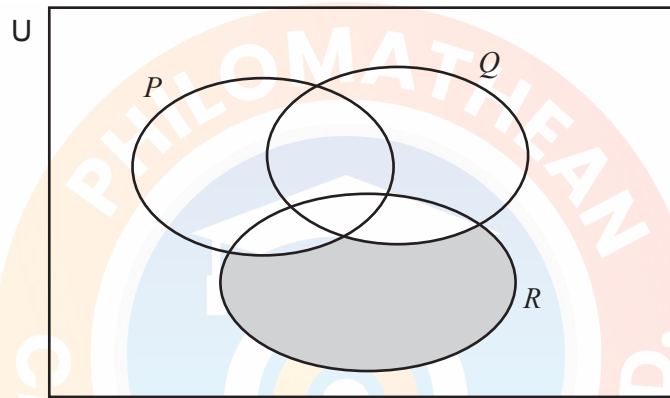
Question 15 is printed on the next page.

- 15 (a) On the Venn Diagram, shade the set $A \cap B \cap C'$.



[1]

- (b) Use set notation to describe the shaded region.



[1]

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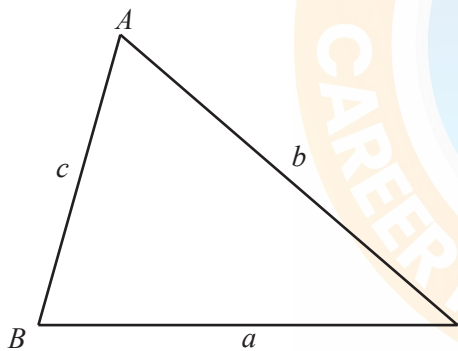
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1 Work out.

(a) $3 - 0.018$

..... [1]

(b) 0.04^2

..... [1]

(c) $\frac{0.08}{0.2}$

..... [1]

2 (a) Write 5249.6 correct to two significant figures.

..... [1]

(b) Write 0.0030626 correct to three decimal places.

..... [1]

3 A car travels 300 metres in 20 seconds.

Find the average speed of the car in

(a) metres per second,

..... m/s [1]

(b) kilometres per hour.

..... km/h [2]

4 Solve.

(a) $2 - 4(5 - 2x) = 0$

$x = \dots\dots\dots$ [2]

(b) $|2x - 5| = 9$

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

5 Find the value of

(a) 64^0 ,

(b) $64^{\frac{1}{3}}$.

$\dots\dots\dots$ [1]

$\dots\dots\dots$ [1]

6 A regular polygon has 30 sides.

Find the size of one exterior angle.

$\dots\dots\dots$ [2]

7 Factorise.

(a) $12ax - 2by + 3ay - 8bx$

..... [2]

(b) $5x^2 - 6x - 8$

8 (a) Work out $\begin{pmatrix} 12 \\ -5 \end{pmatrix} - 5\begin{pmatrix} 4 \\ -1 \end{pmatrix}$.

..... [2]

$\begin{pmatrix} \\ \end{pmatrix}$ [2]

(b) Work out the magnitude of $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

..... [2]

- 9 Rearrange this equation to make x the subject.

$$\frac{a}{2x-3} = \frac{b}{5x}$$

$$x = \dots\dots\dots [3]$$

- 10 (a) Solve.

$$\sin x = \frac{1}{2} \text{ for } 0^\circ \leq x \leq 90^\circ$$

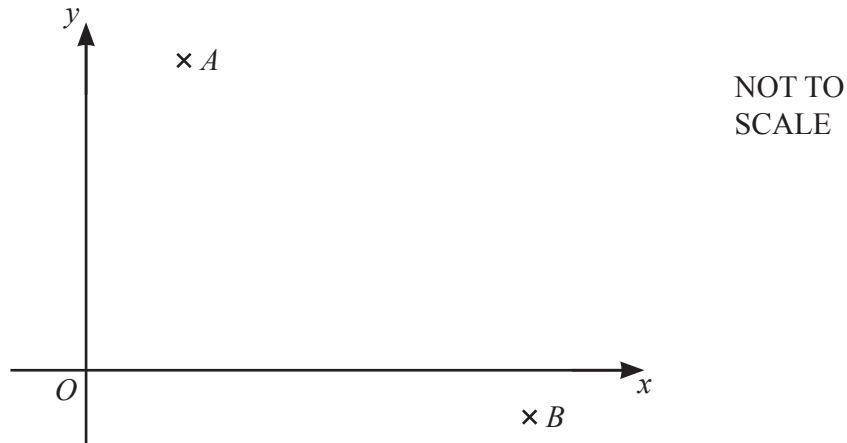
$$x = \dots\dots\dots [1]$$

- (b) Solve.

$$\sin x = -\frac{1}{2} \text{ for } 0^\circ \leq x \leq 360^\circ$$

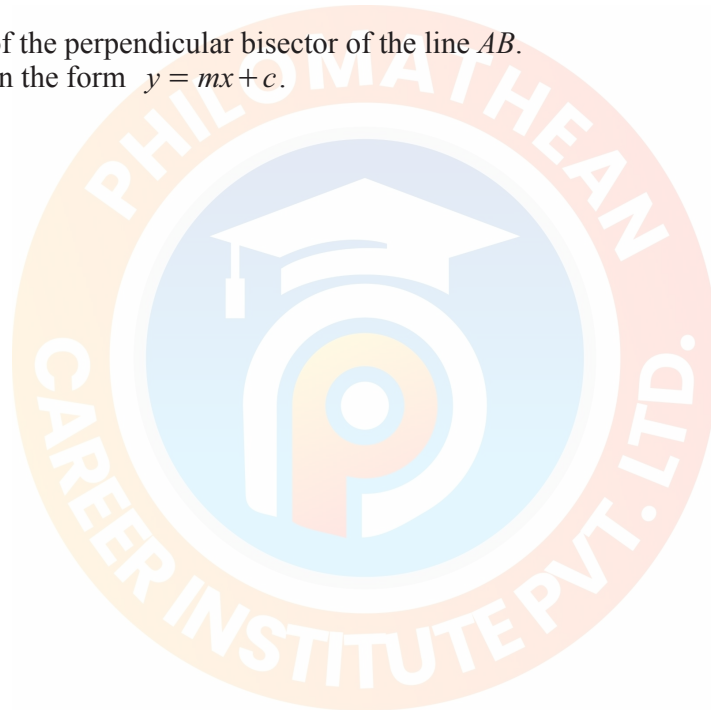
$$x = \dots\dots\dots [2]$$

11



The points $A(2, 8)$ and $B(6, -2)$ are shown on the diagram.

Find the equation of the perpendicular bisector of the line AB .
Give your answer in the form $y = mx + c$.



$y = \dots\dots\dots$ [5]

Question 12 is printed on the next page.

- 12 A bag contains 12 discs.
 7 discs are red and 5 discs are green.
 A disc is picked at random and not replaced.
 A second disc is then picked at random.

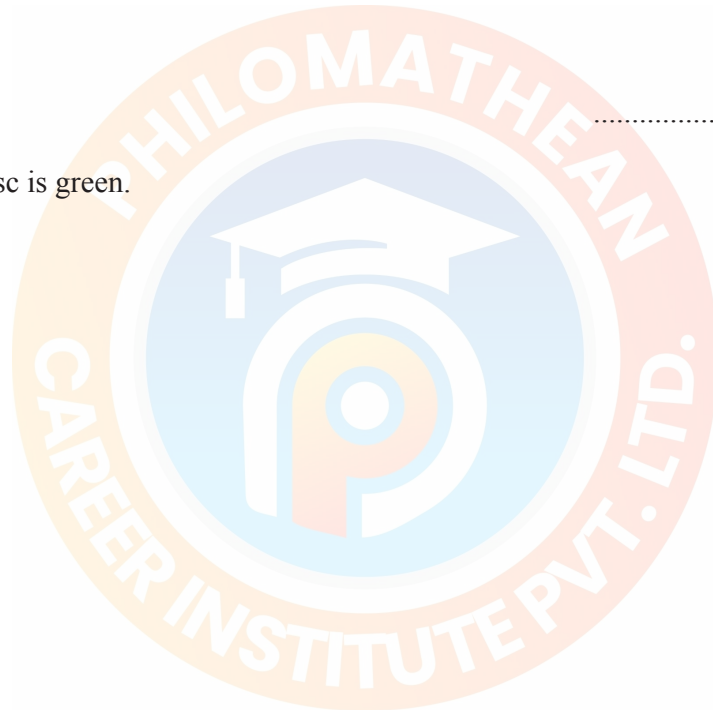
Find the probability that

- (a) both discs are green,

- (b) at least one disc is green.

..... [2]

..... [3]



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CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/22

Paper 2 (Extended)

May/June 2021

45 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly and you will be given marks for correct methods even if your answer is incorrect.
- All answers should be given in their simplest form.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

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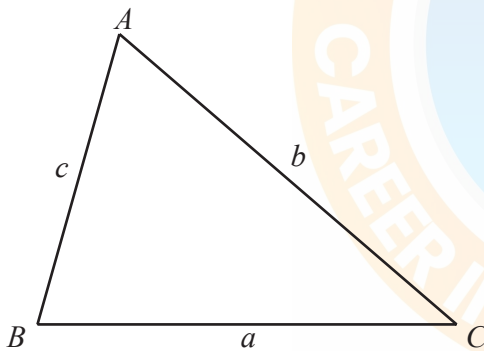
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$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

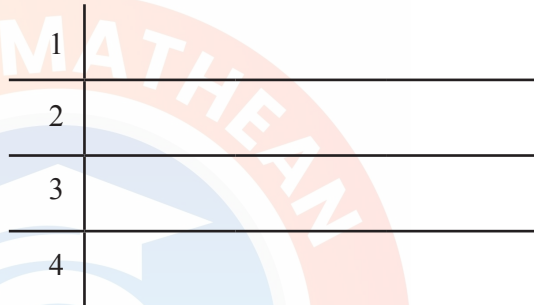
1 Work out $\frac{3.6}{0.004}$.

..... [1]

2 These are the masses, in kilograms, of 16 newborn babies.

2.5 3.2 3.8 3.2 1.9 3.4 1.7 4.1
3.0 2.8 4.0 2.7 3.9 2.7 4.1 3.7

Complete the ordered stem-and-leaf diagram for the masses.



Key: 3 | 2 = 3.2

[2]

3 Work out $2\frac{1}{2} \div 3\frac{1}{4}$.

Give your answer as a fraction in its simplest form.

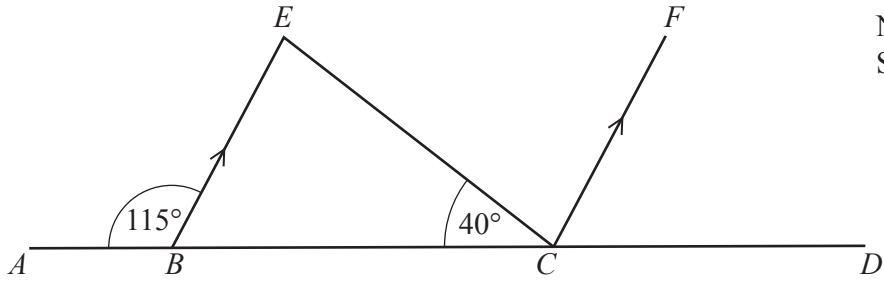
..... [3]

4 Insert **two** pairs of brackets to make this statement correct.

$$3 \times 7 - 3 + 4 \times 2 = 32$$

[1]

5

NOT TO
SCALE

$ABCD$ is a straight line and BE is parallel to CF .

Find angle ECF .

Angle $ECF =$ [2]

6 (a) Factorise $a^2 - b^2$.

..... [1]

(b) Work out $5.37^2 - 4.63^2$.

..... [2]

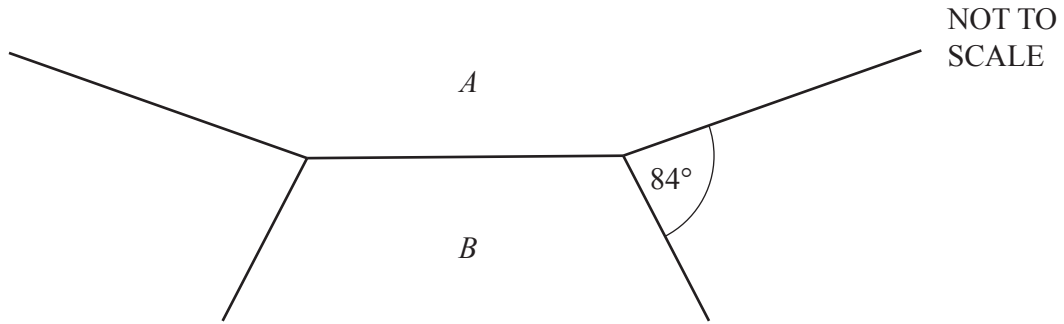
7 Solve $2x + 3 < 5x - 12$.

..... [2]

8 Expand and simplify $(2\sqrt{3} - 5)(4 + \sqrt{3})$.

..... [2]

9



The diagram shows part of polygon A and part of polygon B .

A is a regular polygon with n sides.

B is a regular hexagon.

Find the value of n .

10

$$c = 4 \times 10^7$$

$$d = 5.8 \times 10^6$$

Work out, giving your answers in standard form,

(a) c^2 ,

$$n = \dots\dots\dots [3]$$

(b) $c - d$.

$$\dots\dots\dots [2]$$

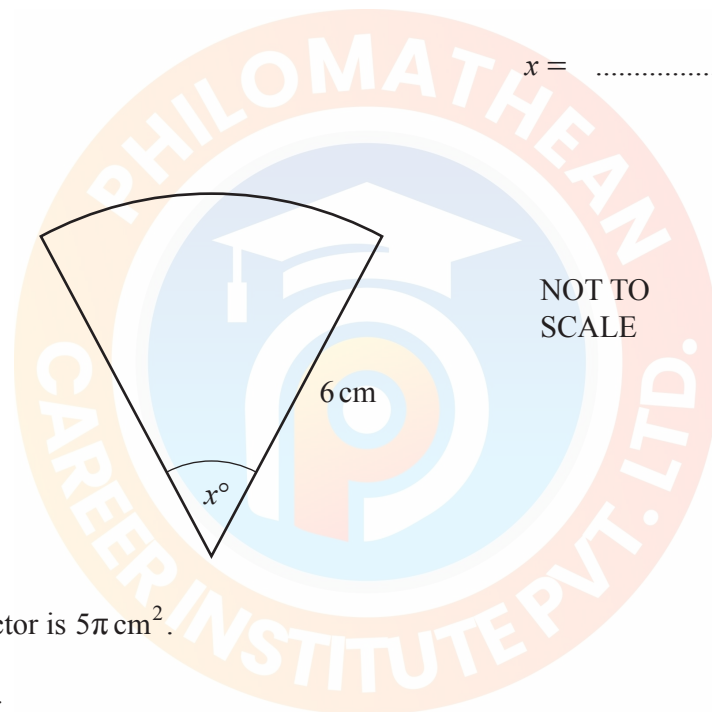
$$\dots\dots\dots [2]$$

11 $y = \frac{2}{x+3}$

Rearrange the formula to make x the subject.

$x = \dots\dots\dots$ [3]

12



The area of this sector is $5\pi \text{ cm}^2$.

Find the value of x .

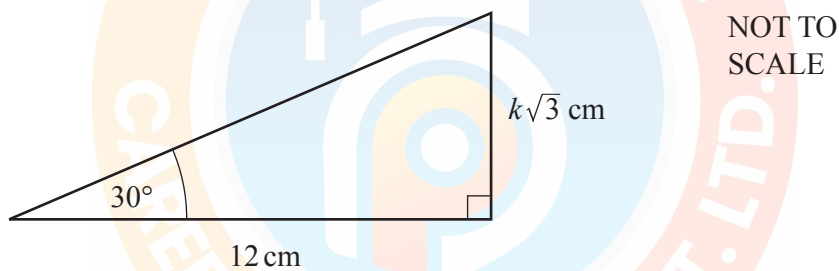
$x = \dots\dots\dots$ [3]

- 13 The heights, h cm, of 100 plants are measured.
The table shows the results.

Height, h cm	Frequency
$0 < h \leq 40$	15
$40 < h \leq 80$	40
$80 < h \leq 120$	45

Calculate an estimate for the mean height of the plants.

14

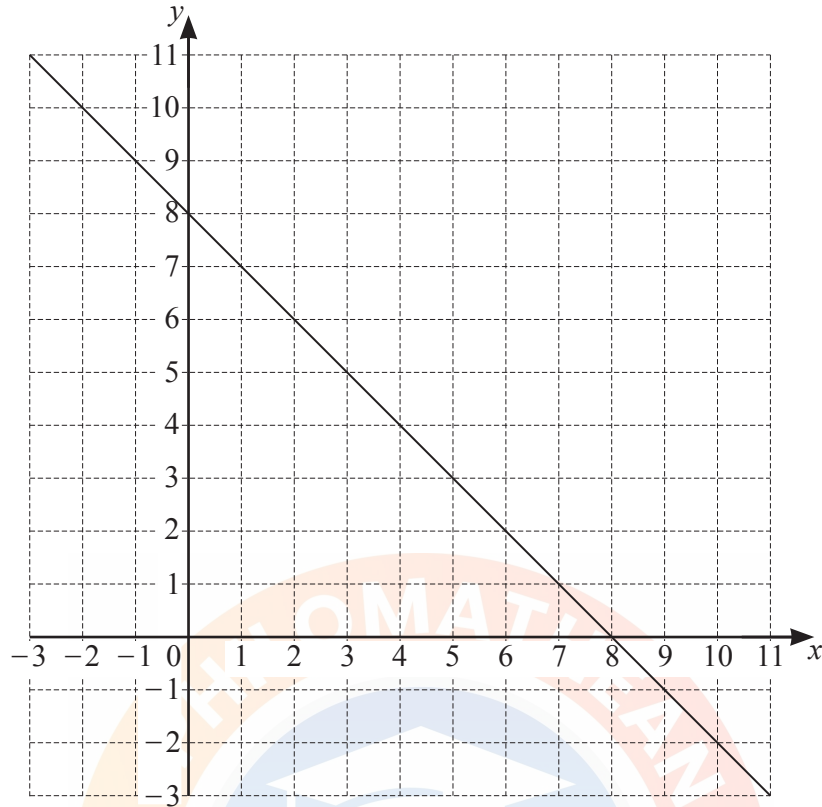


Find the value of k .

$$k = \text{.....} [3]$$

Questions 15 and 16 are printed on the next page.

15



The diagram shows the line $x + y = 8$.

On the diagram, show clearly the region defined by these inequalities.

$$x + y \leq 8$$

$$x \geq 2$$

$$y \leq 3$$

[2]

16 Simplify $\frac{x^2y - 3xy}{x^2 - 2x - 3}$.

[3]

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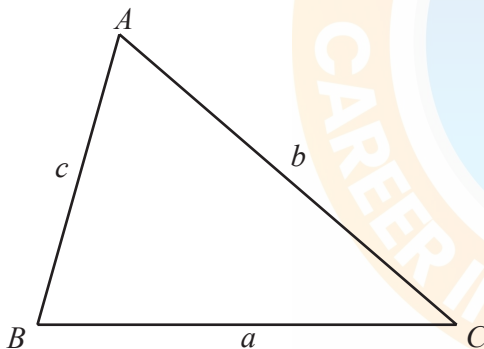
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Volume, V , of cylinder of radius r , height h . $V = \pi r^2 h$

Volume, V , of cone of radius r , height h . $V = \frac{1}{3}\pi r^2 h$

Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 Write 84% as a fraction in its lowest terms.

..... [1]

- 2 Work out $(1 - 0.8)^2$.

..... [1]

- 3 Find the value of $x^2 - x$ when $x = -3$.

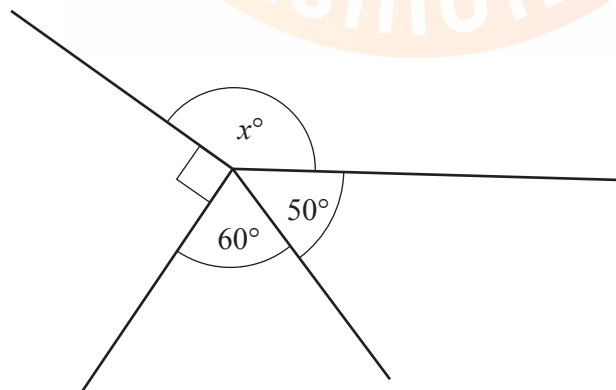
..... [1]

- 4 A quadrilateral has all sides equal and exactly two lines of symmetry.

Write down the mathematical name of this quadrilateral.

..... [1]

5

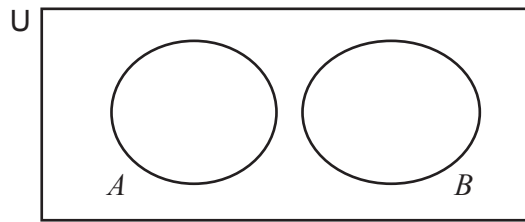


NOT TO
SCALE

Find the value of x .

$x =$ [1]

- 6 On the Venn diagram, shade $A \cup B$.



[1]

- 7 Find the size of one interior angle of a regular polygon with 20 sides.

- 8 Find the value of $|-4| + 4$.

[3]

[1]

- 9 A van has length 9 m.
It takes 1 second for the van to completely pass a gate of length 1 m.

Find the speed of the van.
Give your answer in km/h.

..... km/h [2]

- 10 The faces of a die are numbered 1, 1, 2, 3, 3 and 4.
When it is rolled it is equally likely to show any face.
The die is rolled twice.

Find the probability that it shows an odd number both times.

..... [2]

- 11 Here are the first five terms of a sequence.

$$\frac{1}{4} \quad 1 \quad 4 \quad 16 \quad 64$$

- (a) Find the next term.

..... [1]

- (b) Find the n th term.

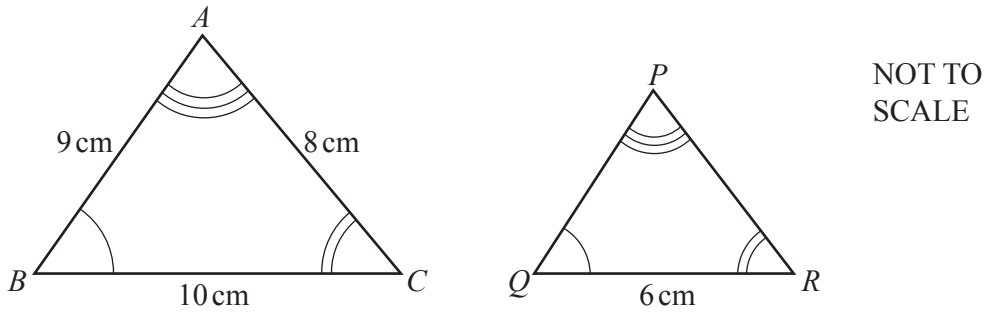
..... [2]

- 12 Factorise.

$$1 + a - c - ac$$

..... [2]

13



The diagram shows two similar triangles, ABC and PQR .

- (a) Find the length of PR .

$PR = \dots\dots\dots\text{ cm}$ [2]

- (b) The triangles are the cross-sections of mathematically similar prisms. The volume of the larger prism is 500 cm^3 .

Find the volume of the smaller prism.

$\dots\dots\dots\text{ cm}^3$ [2]

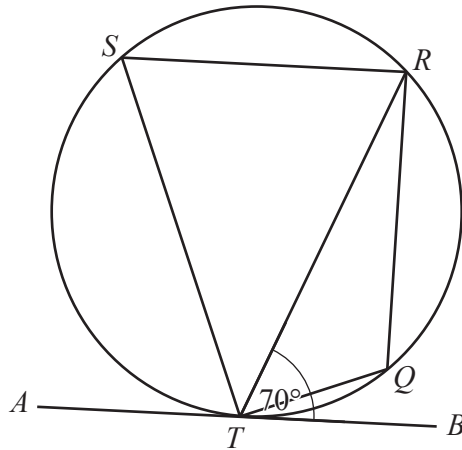
14

$$A = P(1+x)^3$$

Rearrange the formula to write x in terms of A and P .

$x = \dots\dots\dots$ [3]

15

NOT TO
SCALE

Points Q , R , S and T lie on the circle.
 AB is a tangent to the circle at T .
 Angle $RTB = 70^\circ$.

Find angle RQT .

Angle $RQT = \dots\dots\dots$ [2]

- 16 p varies inversely as the square root of q .
 When $q = 9$, $p = 12$.

Find p when $q = 16$.

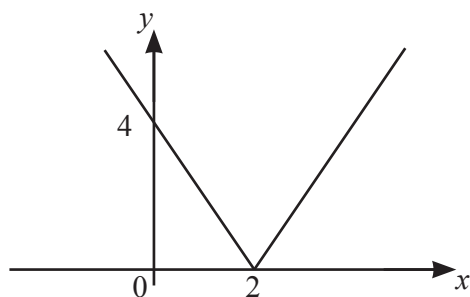
$p = \dots\dots\dots$ [3]

- 17 Simplify by rationalising the denominator. $\frac{3}{2\sqrt{2}-1}$

$\dots\dots\dots$ [2]

Questions 18, 19 and 20 are printed on the next page.

18

NOT TO
SCALE

The diagram shows the graph of $y = |ax + b|$, where $a > 0$.

Find the value of a and the value of b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots [2]$$

19 Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - 2$$

$$\dots\dots\dots [2]$$

20 $2 \log p = 3 \log x - \log y$

Find p in terms of x and y .

$$p = \dots\dots\dots [3]$$

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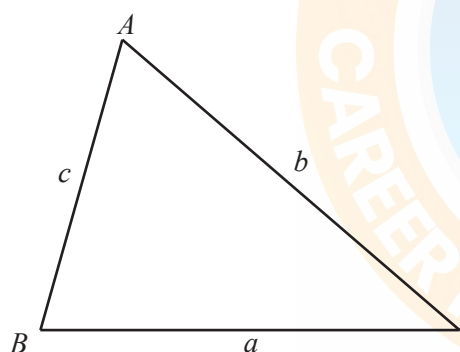
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



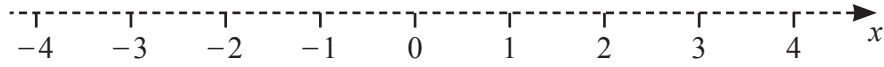
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 On the number line, show the inequality $-2 \leq x < 3$.



[2]

- 2 Work out $4 \times \begin{pmatrix} 6 \\ -2 \end{pmatrix}$.

$\begin{pmatrix} \\ \end{pmatrix}$ [1]

- 3 21 24 25 27 29 39 48

From the list of numbers, write down

- (a) the prime number,

..... [1]

- (b) the cube number.

..... [1]

- 4 Factorise $x^3 - 2x$.

..... [1]

- 5 (a) Write 7.29784 correct to 3 significant figures.

..... [1]

- (b) Write 0.00000306 in standard form.

..... [1]

- 6 Solve.

- (a) $4x = 28$

$x =$ [1]

- (b) $3(a - 6) = 24$

$a =$ [2]

- 7 Karen has 3 blue hats, 5 red hats and 2 white hats.
She also has 4 blue scarves, 3 red scarves and 1 white scarf.

(a) Karen takes a hat at random and replaces it.

Find the probability that it is white.

..... [1]

(b) Karen takes a hat and a scarf at random.

Find the probability that both the hat and the scarf are blue.

..... [2]

- 8 Find the value of $49^{\frac{1}{2}}$.

..... [1]

- 9 Write 90 as the product of its prime factors.

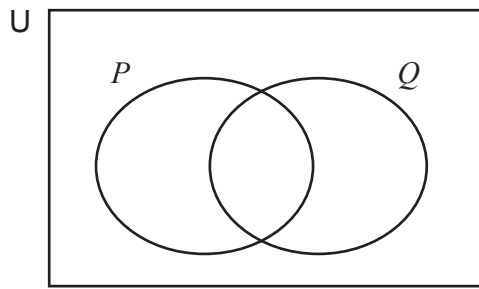
..... [2]

- 10 Find the magnitude of the vector $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$.

Give your answer in simplest surd form.

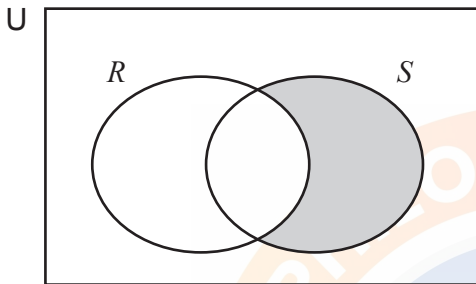
..... [2]

- 11 (a) Shade $P \cup Q$.



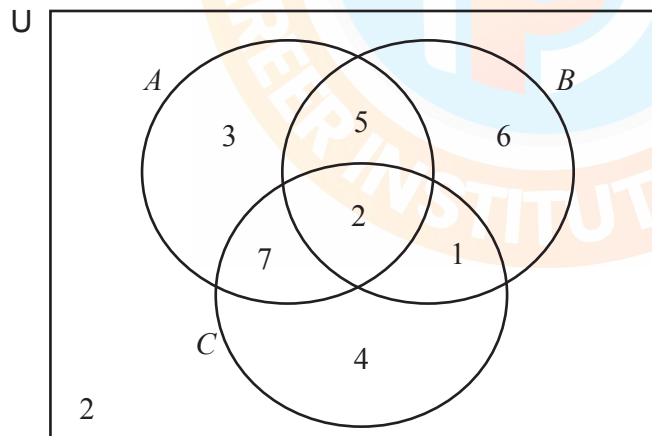
[1]

- (b) Describe the shaded area using set notation.



[1]

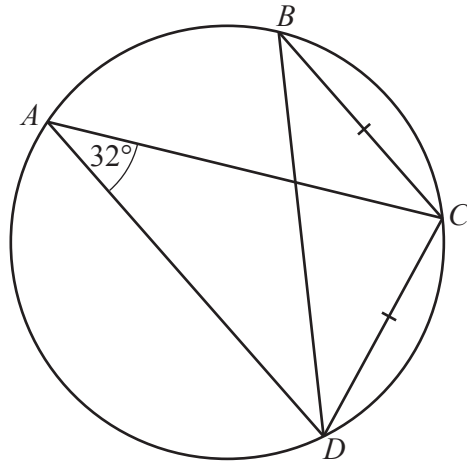
- (c) The Venn diagram shows the number of elements in each subset.



Find $n((B' \cap C) \cap A)$.

[1]

12 (a)

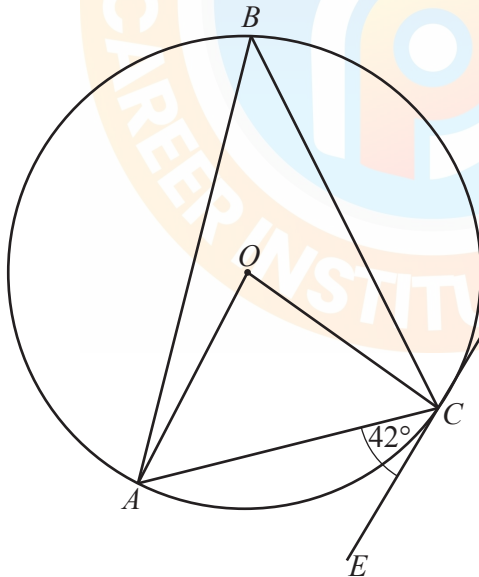
NOT TO
SCALE

$A, B, C,$ and D are points on a circle.
 Angle $DAC = 32^\circ$.
 $BC = DC$.

Find angle BCD .

Angle $BCD = \dots\dots\dots$ [2]

(b)

NOT TO
SCALE

A, B and C are points on the circle centre O .
 ECD is a tangent to the circle at C .
 Angle $ACE = 42^\circ$.

Find angle AOC .

Angle $AOC = \dots\dots\dots$ [2]

- 13 (a) Simplify fully.

$$\sqrt{75} - \sqrt{48} + \sqrt{12}$$

..... [2]

- (b) Rationalise the denominator, giving your answer in its simplest form.

$$\frac{1}{\sqrt{3} + 5}$$

..... [2]

14

$$x^2 - 14x + c = (x + d)^2$$

Find the value of c and the value of d .

$c =$

$d =$ [3]

Questions 15 and 16 are printed on the next page.

15 (a) Factorise fully.

$$6x^2 - 7x - 3$$

..... [2]

(b) Solve.

$$6x^2 - 7x - 3 < 0$$

..... [3]

16 Solve.

$$2\log 3 - \log 2 = \log p$$

$p =$ [2]

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CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/22

Paper 2 (Extended)

May/June 2022

45 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly and you will be given marks for correct methods even if your answer is incorrect.
- All answers should be given in their simplest form.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

Formula List

For the equation $ax^2 + bx + c = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

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Curved surface area, A , of cone of radius r , sloping edge l . $A = \pi rl$

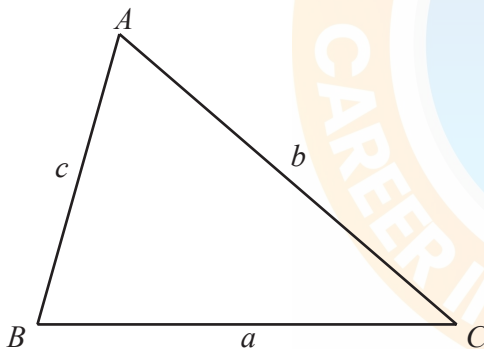
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Volume, V , of cone of radius r , height h . $V = \frac{1}{3}\pi r^2 h$

Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 Work out.

$$(0.03)^2$$

..... [1]

- 2 (a) Write the fraction $\frac{15}{40}$ in its lowest terms.

..... [1]

- (b) Work out.

$$\frac{2}{3} + \frac{2}{9}$$

..... [2]

- 3 Solve the equation.

$$x - 11 = -4$$

$x =$ [1]

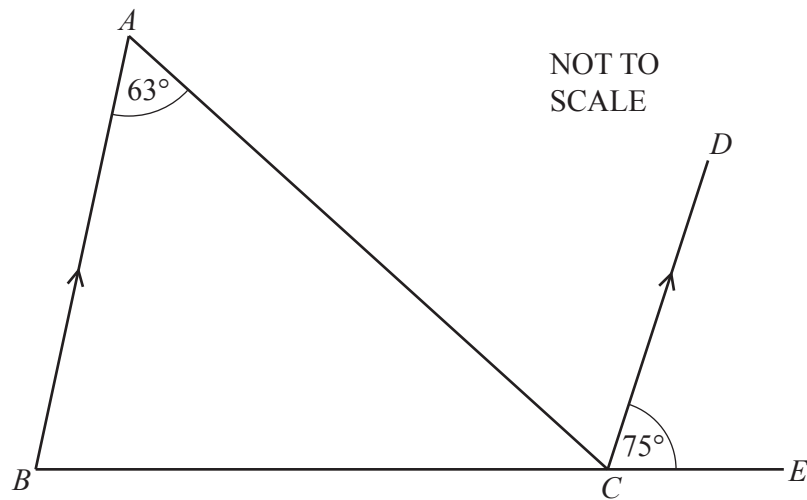
- 4 Change 600 cm^3 into m^3 .

..... m^3 [1]

- 5 Work out $64^{\frac{1}{3}}$.

..... [1]

6



AB is parallel to CD .

Find angle ACD .

Angle $ACD = \dots\dots\dots$ [1]

- 7 Kendra jogs 7 km in 45 minutes.
She then runs at 12 km/h for 30 minutes.
Find her average speed in km/h for the whole journey.

$\dots\dots\dots$ km/h [3]

- 8 The mean of eight numbers is 25.
When two extra numbers are included the mean of the ten numbers is 24.

Find the mean of the two extra numbers.

$\dots\dots\dots$ [2]

- 9 Solve the simultaneous equations.

$$5x + 2y = -12$$

$$3x - y = -5$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots [3]$$

- 10 A is the point $(-1, 13)$ and B is the point $(3, 1)$.

Find the equation of the line AB , giving your answer in the form $y = mx + c$.

$$y = \dots\dots\dots [3]$$

- 11 Solve.

$$6x^2 - 5x - 6 = 0$$

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots [3]$$

- 12** The lengths of the sides of a triangle are 3 cm, 4 cm and 5 cm.

Find the sine of the smallest angle.

..... [1]

- 13** John goes to a shop that sells newspapers and magazines only.

(a) Complete the table of probabilities of John buying something at the shop.

	Buys a newspaper	Does not buy a newspaper	Total
Buys a magazine			0.40
Does not buy a magazine	0.25		
Total	0.55		1.00

[2]

(b) Find the probability that John buys a magazine but not a newspaper.

..... [1]

- 14** $f(x) = |2x + 3|$

Find the values of x when $f(x) = 15$.

..... [2]

- 15** A bag has 5 black counters, 4 white counters and 1 red counter.
One counter is chosen at random and is replaced.
A second counter is then chosen at random.

Find the probability that the two counters chosen are different colours.

..... [4]

- 16** Solve.

$$\log x = 1 + \log 9 - \log 8 + 2 \log \frac{2}{3}$$

$x =$ [3]

Question 17 is printed on the next page.

17 (a) Expand the brackets and simplify.

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})$$

..... [2]

(b) Rationalise the denominator.

$$\frac{1}{\sqrt{7} + \sqrt{6}}$$

..... [1]

(c) Work out the value of

$$\frac{1}{\sqrt{9} + \sqrt{8}} + \frac{1}{\sqrt{8} + \sqrt{7}} + \frac{1}{\sqrt{7} + \sqrt{6}} + \frac{1}{\sqrt{6} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{4}}.$$

..... [2]

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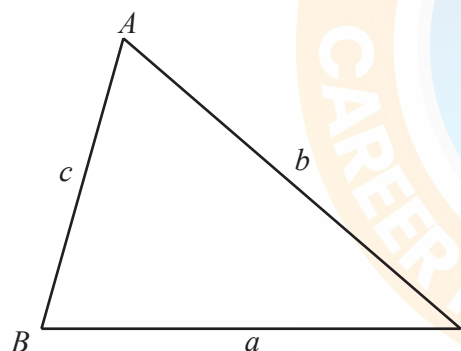
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$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1 Work out.

(a) 0.3×0.2

..... [1]

(b) $12 \div 0.4$

..... [1]

2 This is a list of 8 numbers.

11 7 8 13 7 14 15 5

(a) Find the median.

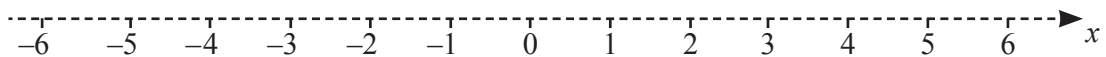
..... [2]

(b) An extra number is added to the list.
The mean of the nine numbers is 1 more than the mean of the eight numbers.
Find the ninth number.

..... [3]

3 Show this inequality on the number line.

$$-3 < x \leq 4$$



[2]

- 4 (a) Express 175 as the product of its prime factors.

..... [2]

- (b) Kurt has two timers.
One is set to ring every 175 minutes.
The other is set to ring every 70 minutes.

Both timers ring together at 09 15.

Find the time when the timers next ring together.

..... [3]

- 5 Expand.
 $3(2x - 1)$

..... [1]

- 6 Find the exterior angle of a regular polygon with 15 sides.

..... [2]

- 7 Eggs are graded into four sizes: extra large, large, medium and small. A farmer records the sizes of a sample of 100 eggs that she collects. The results are shown in the table.

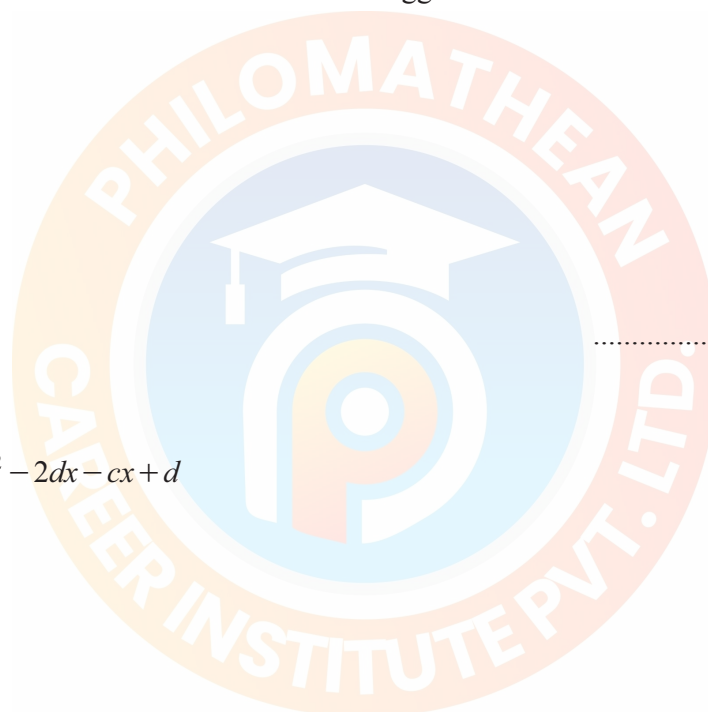
Size	Extra large	Large	Medium	Small
Number of eggs	28	36	24	12

- (a) Find the relative frequency for large eggs.

..... [1]

- (b) In one month, the farmer collects 2500 eggs.

Calculate an estimate for the number of these eggs that are small.

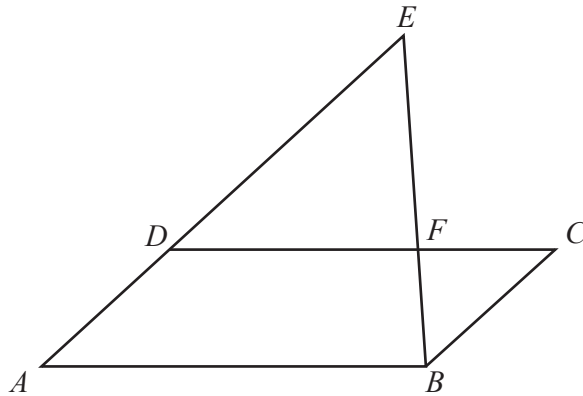


..... [2]

- 8 Factorise fully.

$$2cx^2 - 2dx - cx + d$$

..... [2]



NOT TO
SCALE

$ABCD$ is a parallelogram.
 EDA and EFB are straight lines.

- (a) Show that triangles EDF and BCF are similar.

[2]

- (b) $BC = 4$ cm, $DE = 5$ cm and $FB = 3$ cm.

Find EF .

$EF = \dots\dots\dots$ cm [2]

10 A is the point $(-5, 7)$ and C is the point $(1, -2)$.

(a) B is the mid-point of AC .

Find the coordinates of B .

(.....,) [2]

(b) The line CD is perpendicular to the line AC .

Find the equation of line CD .

11 y is inversely proportional to $(x+2)^2$.
When $x = 3$, $y = 2$.

(a) Find y in terms of x .

$y =$ [2]

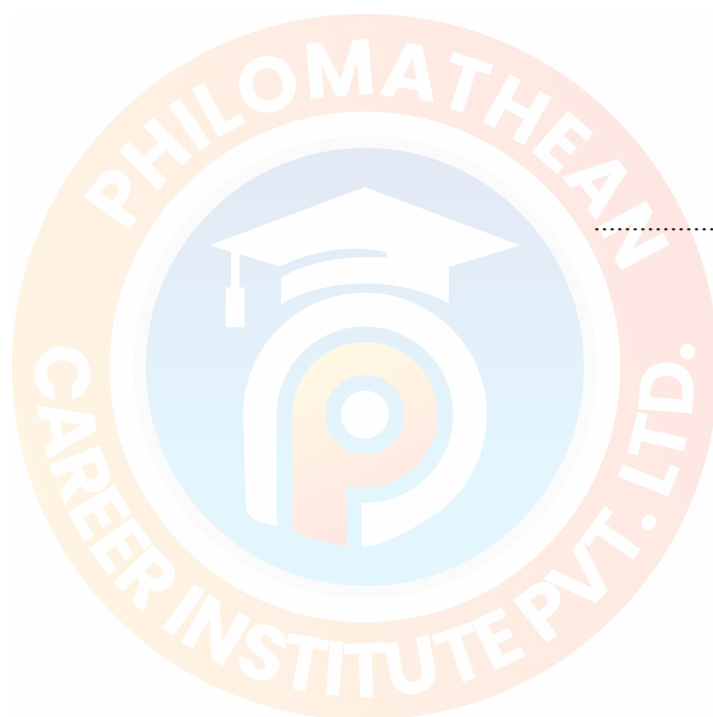
(b) Find the positive value of x when $y = 0.5$.

$x =$ [2]

Question 12 is printed on the next page.

12 $\mathbf{a} = \begin{pmatrix} 4 \\ -10 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$

Find the magnitude of the vector $\mathbf{a} - \mathbf{b}$.
Give your answer in its simplest surd form.



[4]

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CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/21

Paper 2 (Extended)

May/June 2023

45 minutes

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- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly and you will be given marks for correct methods even if your answer is incorrect.
- All answers should be given in their simplest form.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

Formula List

For the equation $ax^2 + bx + c = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Curved surface area, A , of cylinder of radius r , height h . $A = 2\pi rh$

Curved surface area, A , of cone of radius r , sloping edge l . $A = \pi rl$

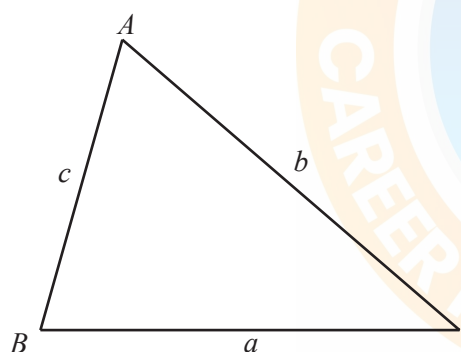
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 (a) Insert **one** pair of brackets to make the statement correct.

$$3 \times 7 + 2 + 9 = 36 \quad [1]$$

- (b) Work out $(0.2)^3$.

..... [1]

- (c) Write down a prime number between 80 and 90.

..... [1]

- 2 Solve the equation.

$$7 - 5x = -3$$

$x =$ [2]

- 3 (a) Work out $\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -5 \\ 3 \end{pmatrix}$.

$\begin{pmatrix} \\ \end{pmatrix}$ [1]

- (b) P is the point $(-3, 6)$.
 Q is the point $(0, 2)$.

Find the translation vector that maps the point P onto the point Q .

$\begin{pmatrix} \\ \end{pmatrix}$ [2]

- 4 (a) Factorise.

$$2p^2 - pq$$

..... [1]

- (b) Expand the brackets and simplify.

$$(p - 7)(p + 3)$$

..... [2]

- 5 (a) Work out $\frac{11}{12} + \frac{3}{4}$.

Give your answer as a mixed number in its simplest form.

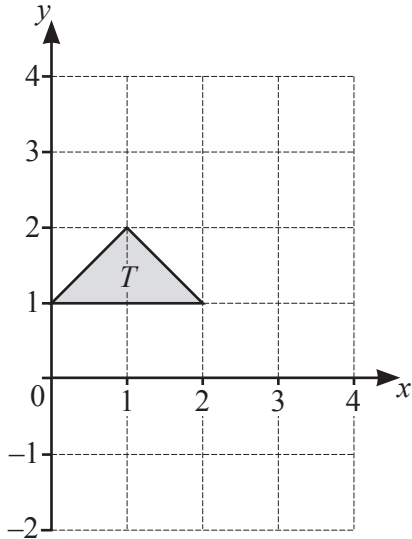
..... [2]

- (b) Simplify $\frac{a}{x} \div \frac{b}{2y}$.

Give your answer as a single fraction.

..... [1]

6



Rotate triangle T 90° clockwise about the point $(2, 1)$.

[2]

7 The interior angle of a regular polygon is 140° .

Find the number of sides of this polygon.

[3]

8 Rearrange this equation to make x the subject.

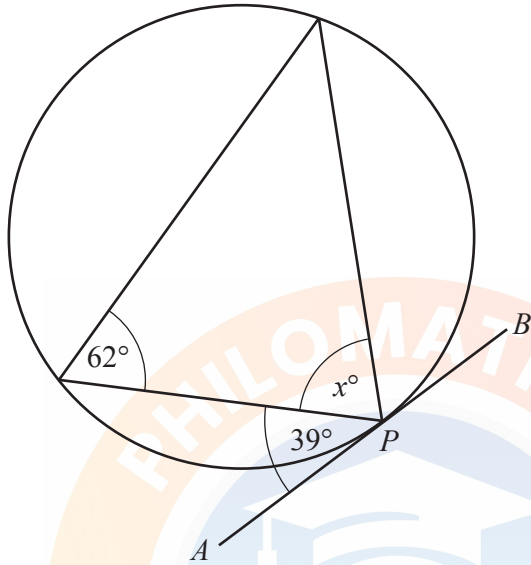
$$y = 7x + 2$$

$x =$ [2]

9 Simplify $(3w^3)^3$.

..... [2]

10



NOT TO
SCALE

APB is a tangent to the circle at P .

Work out the value of x .

$x =$ [2]

11 Simplify $\sqrt{27} + \sqrt{12} - \sqrt{108}$.

..... [2]

12 $f(x) = 3 \sin(4x^\circ)$

Find the amplitude and period of $f(x)$.

Amplitude =

Period = [2]

13 y varies inversely as $\sqrt{x}$.

When $x = 9$, $y = 2$.

Find y in terms of x .

$y = \dots\dots\dots$ [2]

14 $f(x) = x^{\frac{1}{7}}$

Find $f^{-1}(x)$.

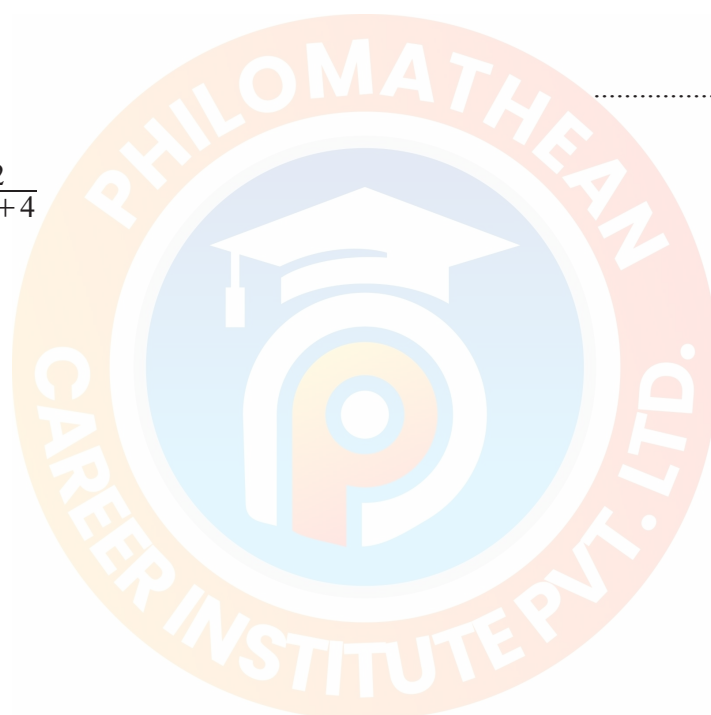
$f^{-1}(x) = \dots\dots\dots$ [1]

Question 15 is printed on the next page.

15 Simplify.

(a) $\frac{3}{x+2} - \frac{2}{x-1}$

(b) $\frac{6x^2 + x - 12}{6ax - 8a - 3x + 4}$



..... [3]

..... [5]

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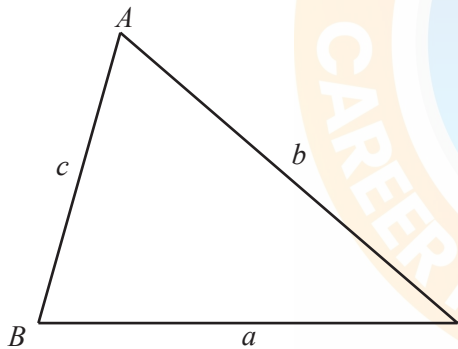
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$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1 Write down

(a) a square number between 101 and 150

..... [1]

(b) a fraction between $\frac{2}{3}$ and $\frac{3}{4}$

..... [1]

(c) an irrational number between 6 and 7.

..... [1]

2 Work out.

(a) $-7 \div -2$

..... [1]

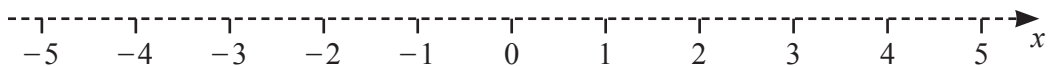
(b) $(0.3)^2$

..... [1]

3 (a) Solve $x + 9 > 6$.

..... [1]

(b) Show your answer to **part (a)** on this number line.



[1]

- 4 Aklima records the masses, m kg, of 120 parcels.
The results are shown in the table.

Mass, m kg	$0 < m \leq 2$	$2 < m \leq 4$	$4 < m \leq 6$	$6 < m \leq 8$	$8 < m \leq 10$
Frequency	35	30	40	12	3

Find

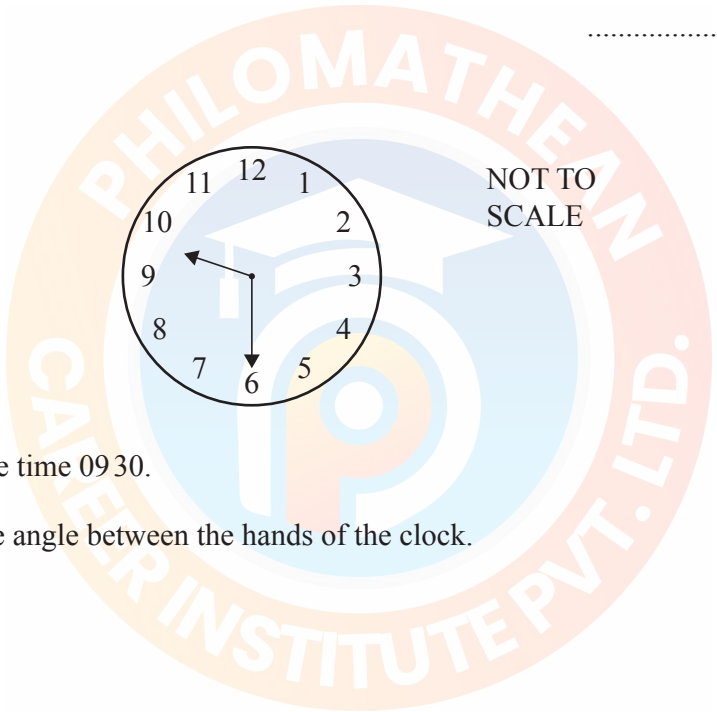
- (a) the modal class

..... $< m \leq$ [1]

- (b) the class which contains the median.

..... $< m \leq$ [1]

5



The clock shows the time 09 30.
Work out the obtuse angle between the hands of the clock.

..... [2]

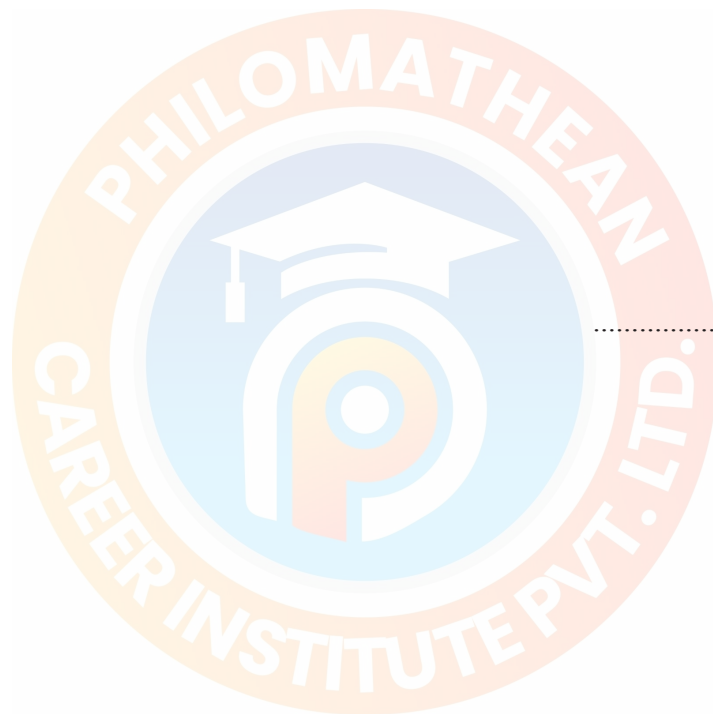
- 6 Find the value of $64^{\frac{1}{3}}$.

..... [1]

- 7 Lee cycles for 60 km at an average speed of 30 km/h.
He then returns along the same route at an average speed of 20 km/h.

Find Lee's average speed for the whole journey.

..... km/h [3]



- 8 Salma spins a biased spinner with sectors labelled 1, 2, 3, 4 and 5. The table shows the relative frequencies of each of her scores.

Score	1	2	3	4	5
Relative frequency	0.1	0.05	0.3	0.35	p

- (a) Find the value of p .

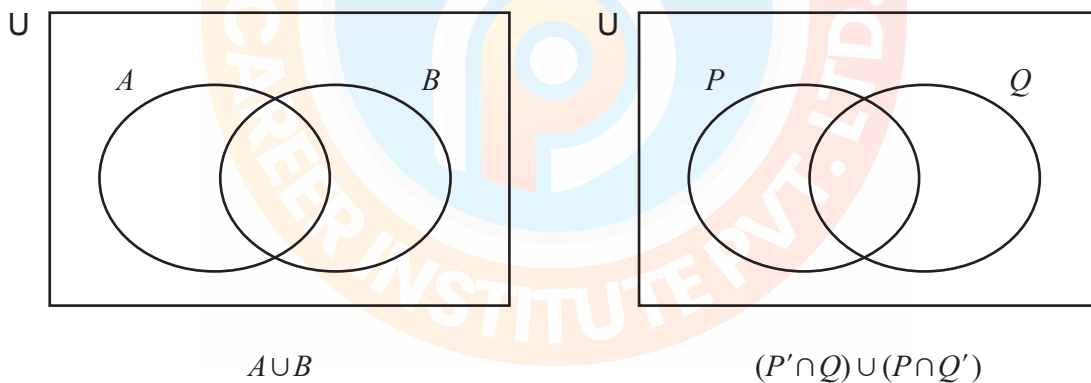
..... [2]

- (b) Salma spins the spinner 4000 times.

Work out an estimate for the number of times she scores 3.

..... [1]

- 9 On the Venn diagrams, shade the given subsets.



[2]

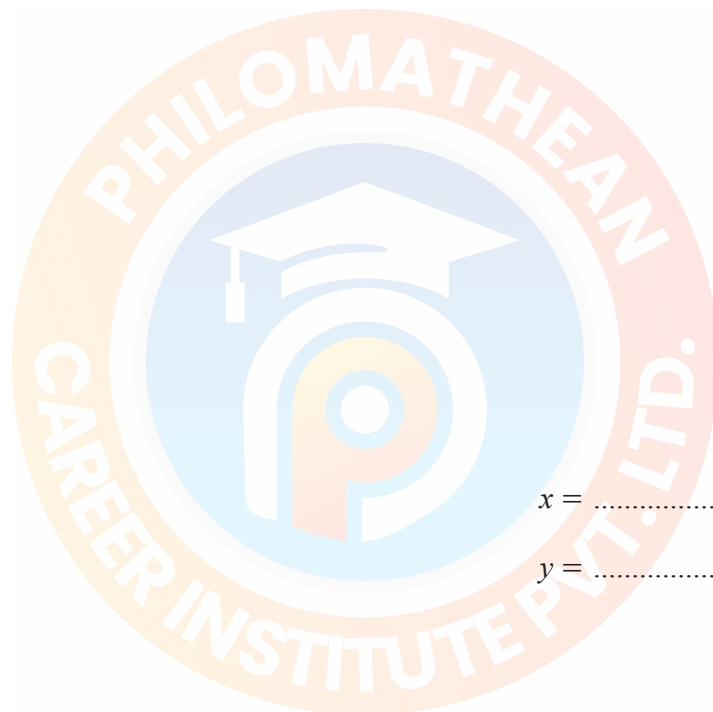
- 10 There is correlation between quantity p and quantity q .
The regression equation is $p = 80 - 5.2q$.

What type of correlation is there between p and q ?

..... [1]

- 11 Solve the simultaneous equations.

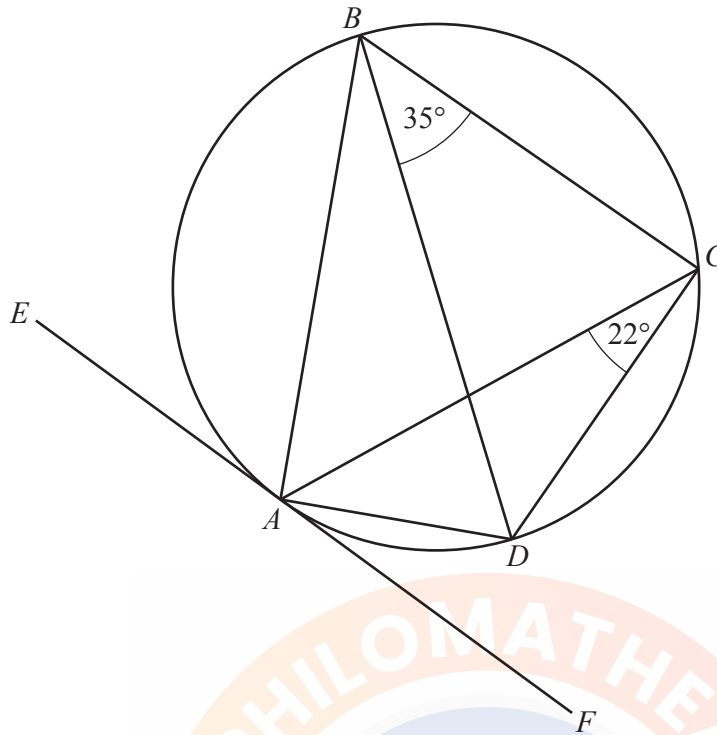
$$\begin{aligned}\frac{1}{2}x - \frac{1}{3}y &= 7 \\ 3x + y &= 6\end{aligned}$$



$x =$

$y =$ [3]

12

NOT TO
SCALE

A, B, C and D are points on the circle.
 EF is a tangent to the circle at A .
 Angle $DBC = 35^\circ$ and angle $ACD = 22^\circ$.

Find

(a) angle ABD Angle $ABD = \dots\dots\dots$ [1](b) angle ADC Angle $ADC = \dots\dots\dots$ [1](c) angle CAF .Angle $CAF = \dots\dots\dots$ [1]

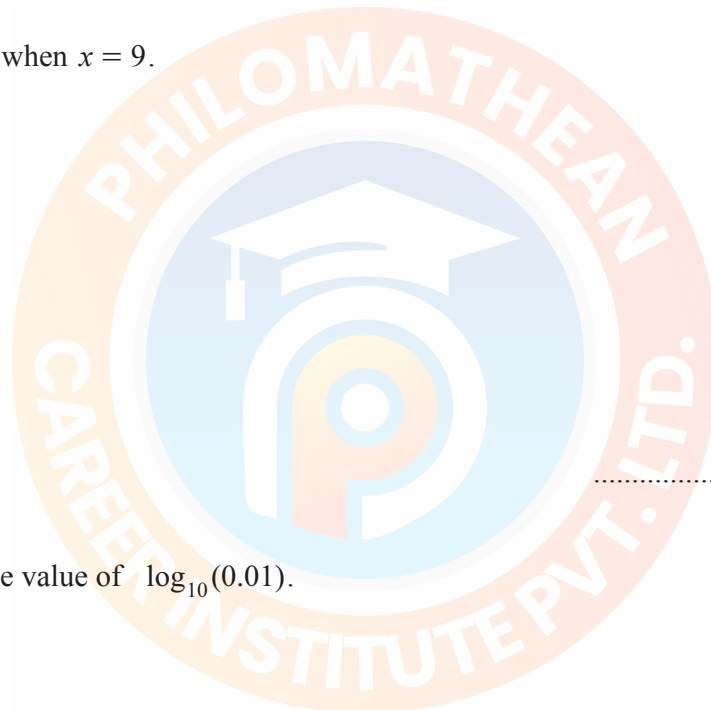
- 13 Rationalise the denominator and simplify.

$$\frac{2}{3 - \sqrt{5}}$$

..... [3]

- 14 y varies inversely as the square of $(x - 3)$.
When $x = 6$, $y = 20$.

Find the value of y when $x = 9$.



..... [3]

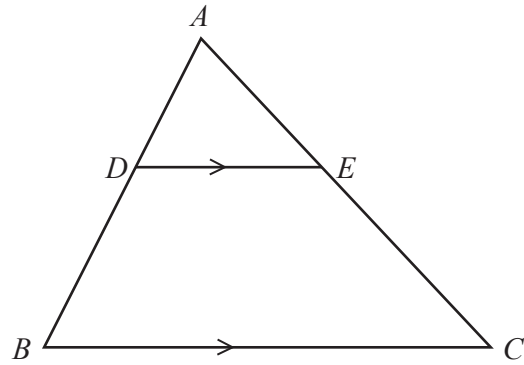
- 15 (a) Write down the value of $\log_{10}(0.01)$.

..... [1]

- (b) Find the value of $2 \log 4 + \log 5 - 3 \log 2$.

..... [3]

16

NOT TO
SCALE

ABC is a triangle.
 DE is parallel to BC .

- (a) Show that triangle ADE is similar to triangle ABC .

.....

.....

.....

.....

..... [2]

- (b) $AD : DB = 2 : 3$.

Find the ratio Area of triangle ADE : Area of triangle ABC .

..... : [1]





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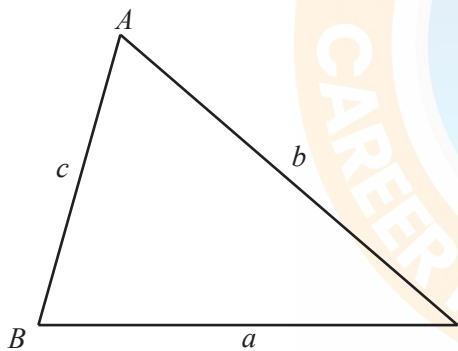
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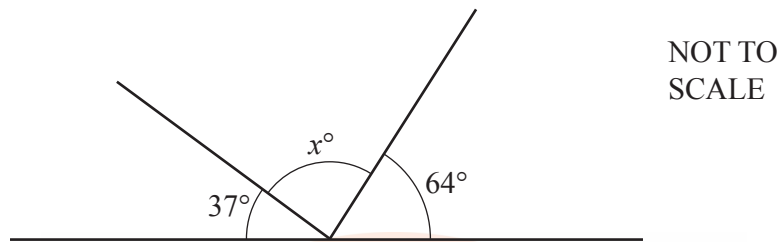
Answer **all** the questions.

- 1 29 31 41 49 51 59

From this list, write down **all** the numbers that are prime numbers.

..... [2]

2



Find the value of x .

$x =$ [1]

3 Work out.

(a) 0.04×0.06

..... [1]

(b) $\frac{0.02}{0.8}$

..... [1]

4 A bag contains 3 blue balls and 7 green balls.
One ball is chosen at random.

Find the probability the ball chosen is

(a) white

..... [1]

(b) not blue.

..... [1]

- 5 Change 12 millimetres into metres.

..... m [1]

- 6 Expand.

$$3x(2x^4 - 5)$$

..... [2]

- 7 (a) Write 0.003 08 in standard form.

..... [1]

- (b) Work out $(7 \times 10^6) \times (3 \times 10^{-8})$.

Give your answer in standard form.

..... [2]

- 8 Find the next term and the n th term for this sequence.

1, 7, 17, 31, 49, ...

next term

n th term [3]

- 9 The total cost of 5 pens and 7 pencils is \$6.75 .
Each pencil costs \$0.45 .

Find the cost of one pen.

\$ [3]

- 10 Write 48 as a product of its prime factors.

..... [2]

- 11 (a) The range of ten numbers is 30.
The range of eight other numbers is 13.

Find the smallest possible value of the range of all eighteen numbers.

..... [1]

- (b) The mean of twelve numbers is 25.
The mean of ten of these numbers is 16.

Find the mean of the other two numbers.

..... [2]

12 Factorise fully.

(a) $(3y)^2 - 16$

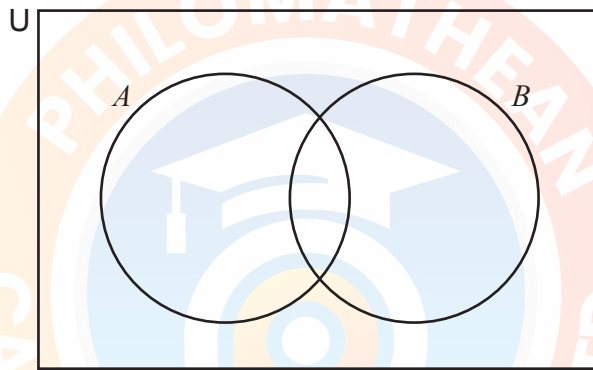
..... [1]

(b) $15ab - 1 - 3a + 5b$

..... [2]

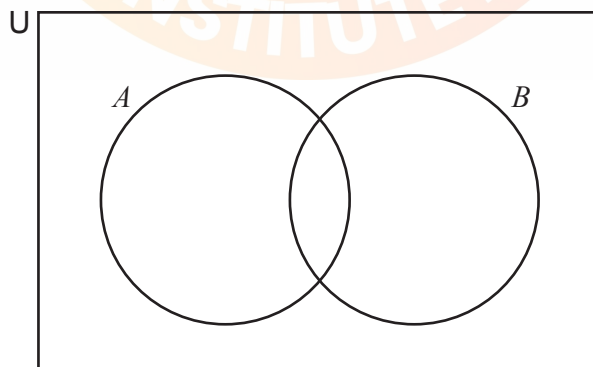
13 Shade the given region on the Venn diagram.

(a) $A' \cap B'$



[1]

(b) $(A \cup B)'$

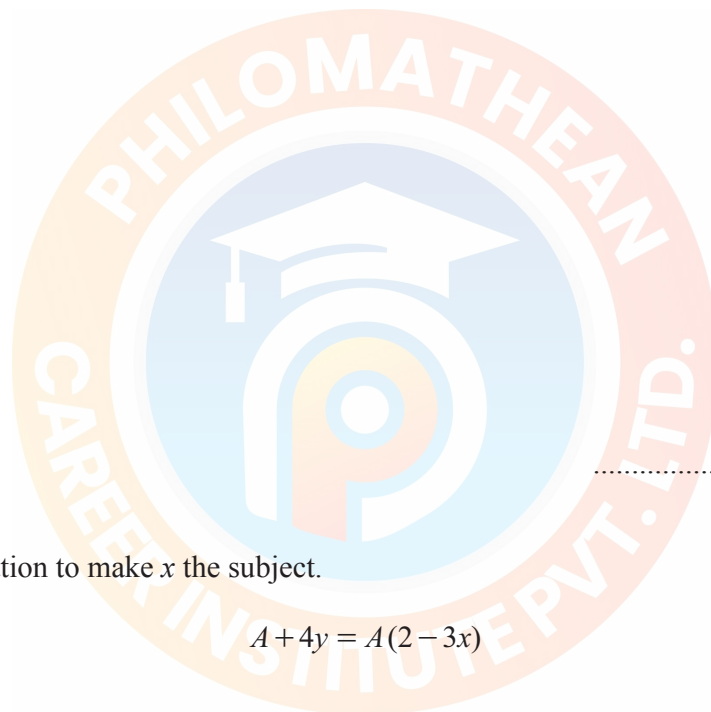


[1]

- 14 An archer shoots three arrows at a target.

The probability that she hits the target with each arrow is $\frac{7}{10}$.

Find the probability she hits the target exactly twice.



- 15 Rearrange the equation to make x the subject.

$$A + 4y = A(2 - 3x)$$

[3]

$x =$ [3]

Question 16 is printed on the next page.

- 16 The point A has coordinates $(2, 3)$ and the point B has coordinates $(6, 5)$.
 The point C lies on the line AB .
 The point D has coordinates $(2, 5.5)$.
 CD is perpendicular to AB .

Find the coordinates of C .



(..... ,) [5]

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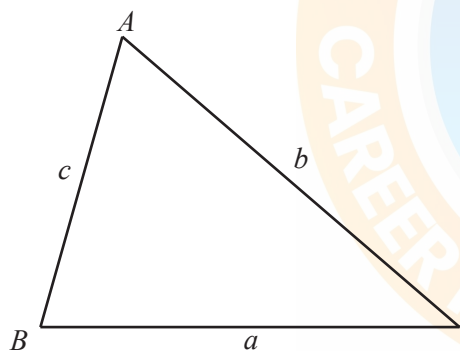
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$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 Work out.

$$2^4$$

..... [1]

- 2 (a) Write $\frac{12}{25}$ as a percentage.

..... % [1]

- (b) Work out.

$$\frac{2}{7} + \frac{4}{7}$$

..... [1]

- 3 Simplify.

$$3x - 2y + x + y$$

..... [2]

- 4 Change 270 mm^2 into m^2 .

..... m^2 [1]

- 5 Write down the value of 9^0 .

..... [1]

- 6 Find the lowest common multiple (LCM) of 24 and 60.

..... [2]

- 7 Find the magnitude of $\begin{pmatrix} 9 \\ -3 \end{pmatrix}$.

Give your answer in its simplest surd form.

- 8 Write in standard form.

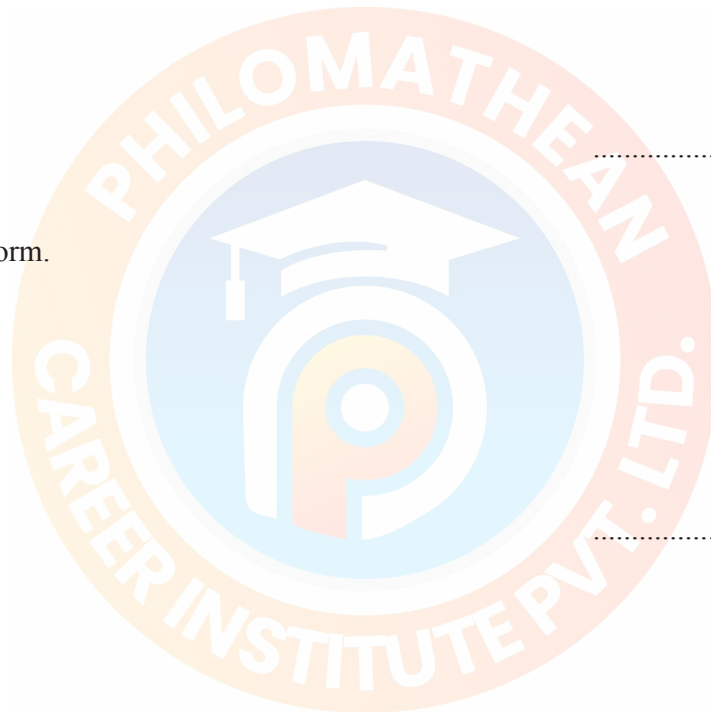
(a) 3 706 000

..... [2]

(b) 0.001 010

..... [1]

..... [1]



- 9 A is the point $(1, 3)$ and B is the point $(3, -7)$.
The line l passes through A and is perpendicular to AB .

Find the equation of line l .

Give your answer in the form $py + qx = r$ where p , q and r are integers.

..... [4]

- 10 The table shows the favourite colour of each of 80 people.

Colour	Red	Blue	White	Green	Silver	Black	Yellow
Frequency	23	17	11	9	12	3	x

- (a) Find the value of x .

..... [1]

- (b) Find the relative frequency of silver, giving your answer as a fraction in its lowest terms.

..... [2]

- 11 (a) Simplify.

$$\sqrt{50} - \sqrt{8}$$

..... [2]

- (b) By rationalising the denominator, simplify

$$\frac{12}{\sqrt{7} - \sqrt{3}}.$$

..... [3]

- 12 Factorise completely.

$$x^3y^2 - xy$$

..... [2]

- 13 Sara has a bag containing 4 yellow balls and 5 white balls.
Sara takes a ball from the bag at random without replacement.
She then takes a second ball from the bag at random.

Find the probability that the two balls are different colours.

..... [3]

14 Make x the subject of the formula.

$$\frac{p}{x} = \frac{q}{x-2}$$

$$x = \dots\dots\dots [3]$$

15 Simplify.

$$1 + 3 \log 2 - 2 \log 3 - 2 \log \frac{2}{3}$$

$$\dots\dots\dots [4]$$

16 Solve the equation.

$$4(\cos x)^2 = 3 \quad \text{for} \quad 0^\circ \leq x \leq 360^\circ$$

$$x = \dots\dots\dots [3]$$



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CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/22

Paper 2 (Extended)

May/June 2024

45 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
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This document has **12** pages. Any blank pages are indicated.



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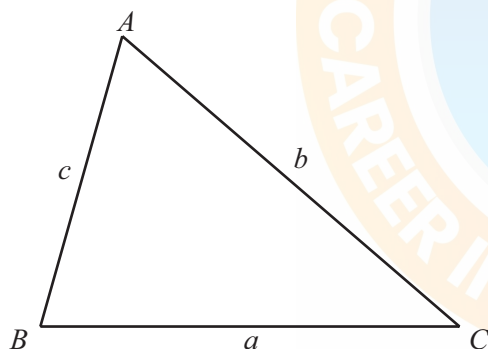
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$$a^2 = b^2 + c^2 - 2bc \cos A$$

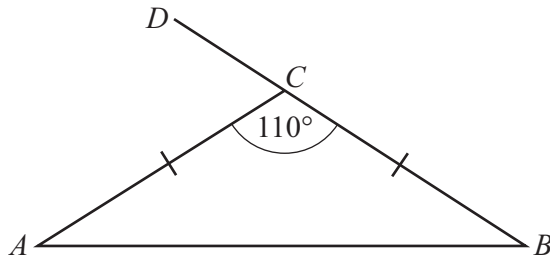
$$\text{Area} = \frac{1}{2}bc \sin A$$





Answer **all** the questions.

1



NOT TO
SCALE

In triangle ABC , $AC = BC$ and BCD is a straight line.

(a) Find angle ACD .

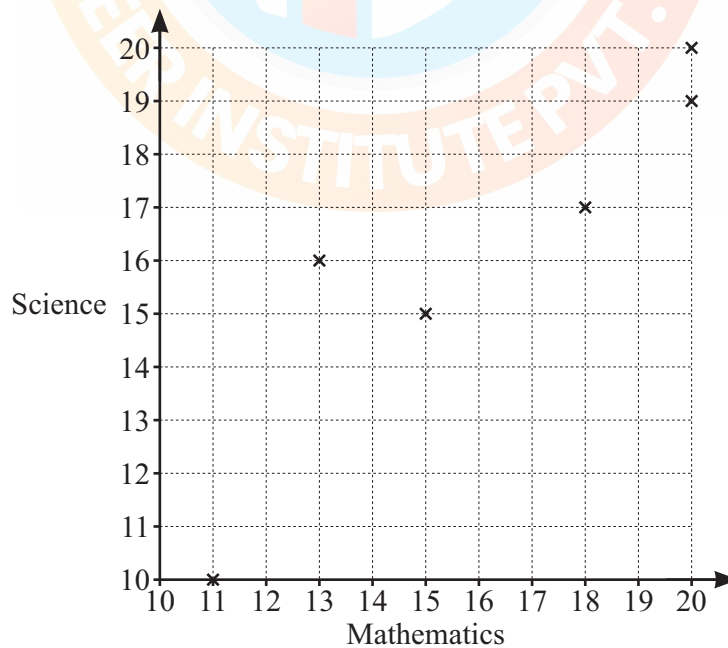
Angle $ACD = \dots\dots\dots$ [1]

(b) Find angle ABC .

Angle $ABC = \dots\dots\dots$ [1]

2 10 students each record their test scores in mathematics and in science.
The table shows the results.

Mathematics	20	18	13	20	11	15	18	14	19	19
Science	20	17	16	19	10	15	20	11	20	16



(a) Complete the scatter diagram.
The first six points have been plotted for you.

[2]

(b) Write down the type of correlation shown in your scatter diagram.





3 Work out 0.4×0.001 .

..... [1]

4 80 81 82 83 84 85 86 87 88 89

From the list of numbers, write down a prime number.

..... [1]

5 Petra repairs cars.

The cost of a repair is a fixed charge of $\$C$ plus a charge of $\$h$ per hour.

Find an expression, in terms of C , h and n , for the total cost, $\$T$, of a repair that takes Petra n hours.

$T =$ [2]

6 Solve the inequality.

$$7 \leq 11x$$

..... [1]





7 (a) Simplify $\sqrt{98}$.

..... [1]

(b) Rationalise the denominator.

$$\frac{3}{\sqrt{5}-2}$$

..... [2]

8 Solve.

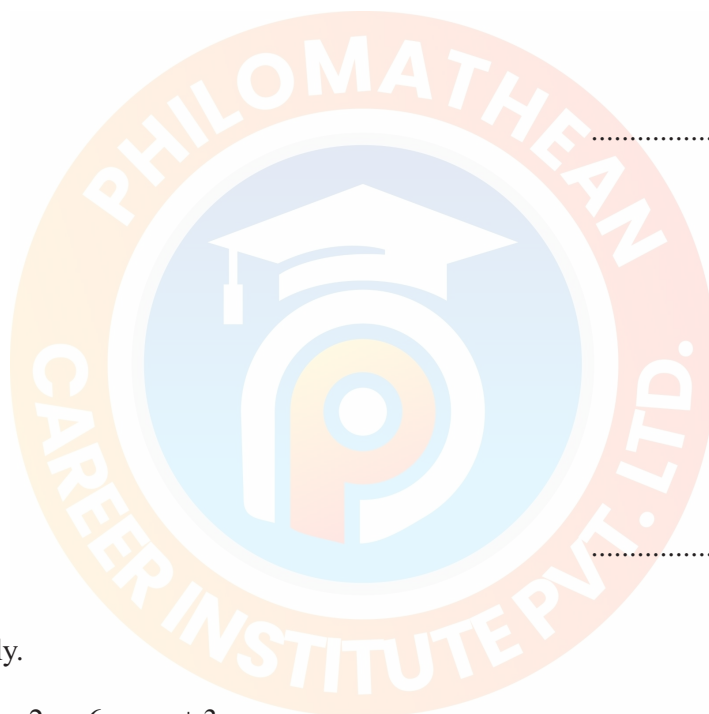
$$|x+11|=7$$

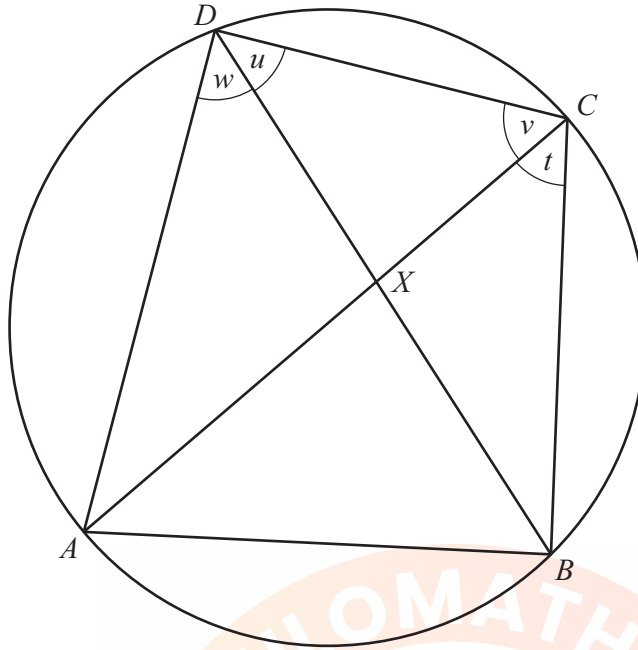
..... [2]

9 Factorise completely.

$$2x-6y-ax+3ay$$

..... [2]





NOT TO
SCALE

$ABCD$ is a cyclic quadrilateral and the diagonals AC and BD intersect at X .

- (a) Complete the statement using two of t , u , v and w .

Angle is equal to angle

[1]

- (b) Angle $DAB = 75^\circ$.

Find angle DCB .

Angle $DCB = \dots\dots\dots$ [1]

- (c) $AB = 8$ cm, $AX = 6$ cm, $BX = 4$ cm and $DC = 5$ cm.

Work out CX .

$CX = \dots\dots\dots$ cm [2]



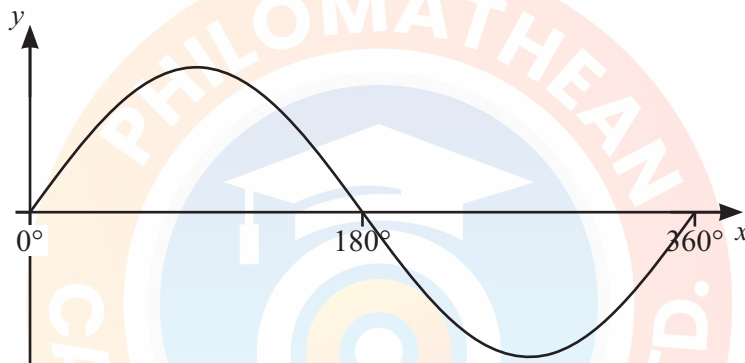


11 $10 < ab < 100$

Simplify $(a \times 10^7) \times (b \times 10^8)$.
Give your answer in standard form.

..... [2]

12



The diagram shows the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$.

(a) On the diagram, sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$. [1]

(b) $\cos x = \sin(x + k)$.

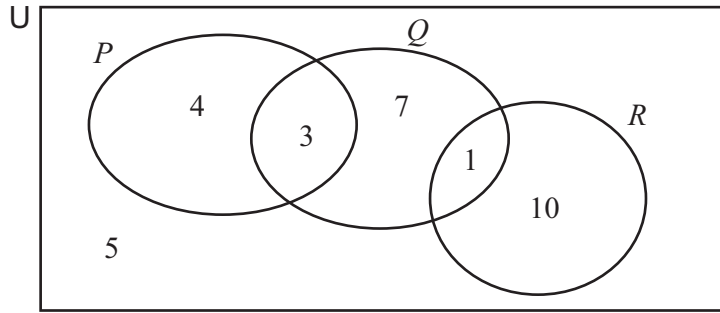
Find a possible value of k .

$k =$ [1]





13



The Venn diagram shows the number of elements in each subset.

(a) Use set notation to complete the statement.

$$P \cap Q \cap R = \dots\dots\dots [1]$$

(b) Find $n((Q \cup R) \cap P')$.

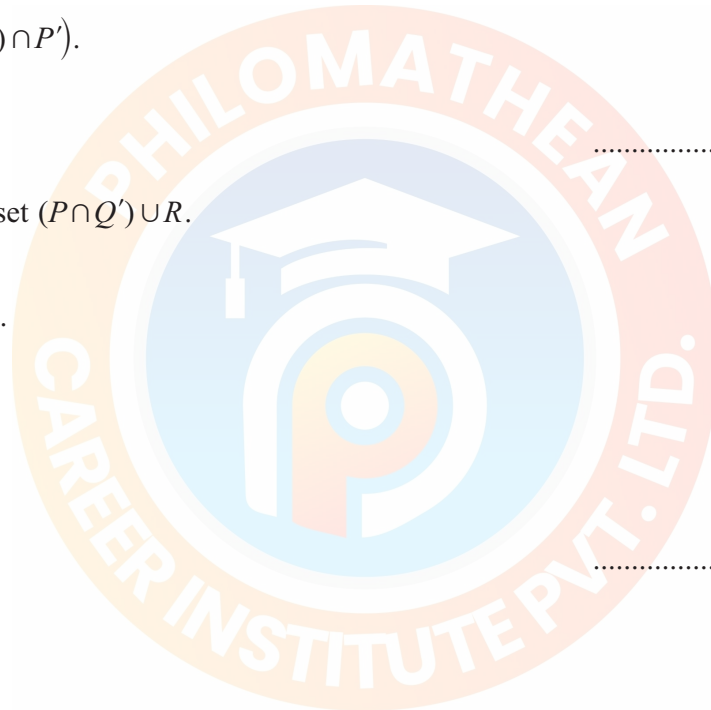
..... [1]

(c) Shade the subset $(P \cap Q') \cup R$.

[1]

14 Simplify $(25x^{16})^{\frac{3}{2}}$.

..... [2]



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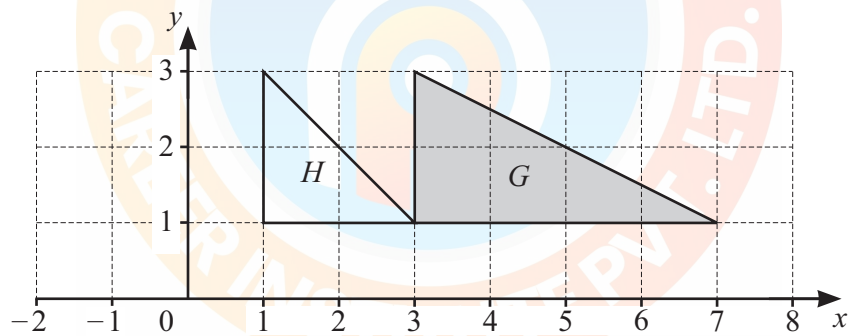




15 A triangle has sides of length 3 cm, 5 cm and 7 cm.

Work out the largest angle in the triangle.

16



Describe fully the **single** transformation that maps shape *G* onto shape *H*.

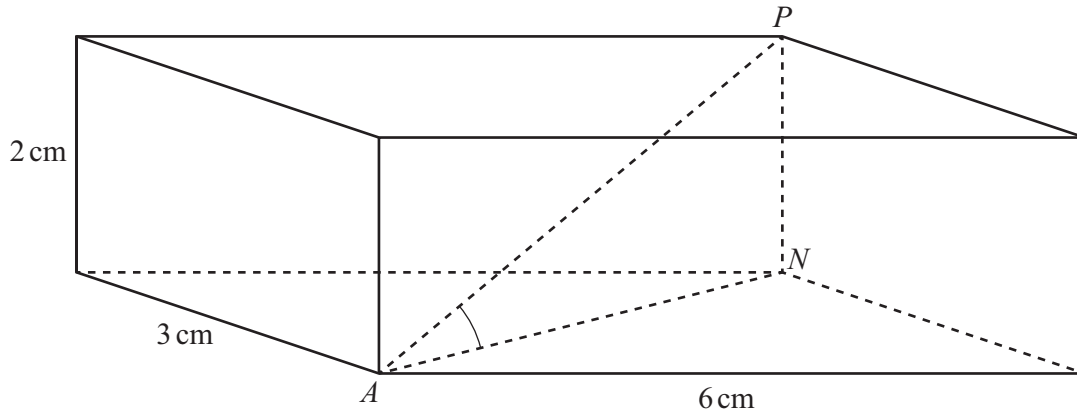
.....

..... [3]





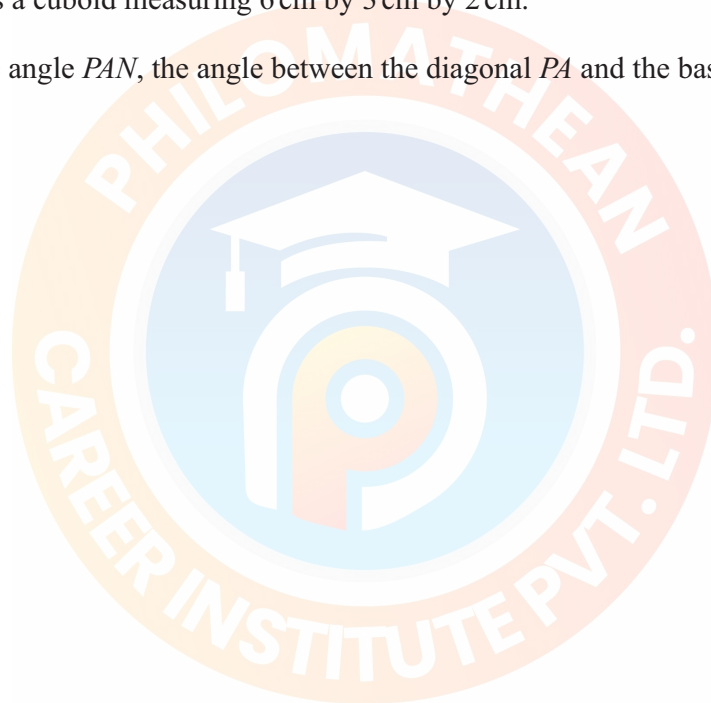
17



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The diagram shows a cuboid measuring 6 cm by 3 cm by 2 cm.

Find the sine of the angle PAN , the angle between the diagonal PA and the base of the cuboid.



Sine of angle PAN = [4]





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Formula List

For the equation $ax^2 + bx + c = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Curved surface area, A , of cylinder of radius r , height h . $A = 2\pi rh$

Curved surface area, A , of cone of radius r , sloping edge l . $A = \pi rl$

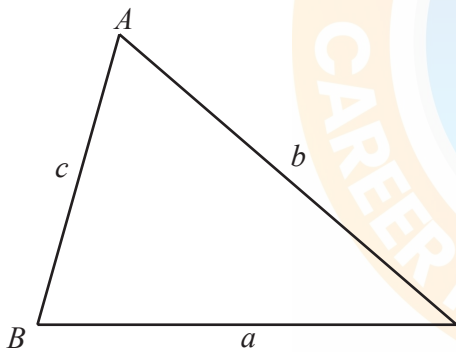
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Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

1 Work out $1.1 - 0.2^2$.

..... [2]

2 Work out $\frac{3}{4} - \frac{1}{6}$.

..... [2]

3 A quadrilateral has rotational symmetry of order 2 and no lines of symmetry.

Write down the mathematical name of this quadrilateral.

..... [1]

4 Simplify fully.

(a) $3a - 6b - 2a + b$

..... [2]

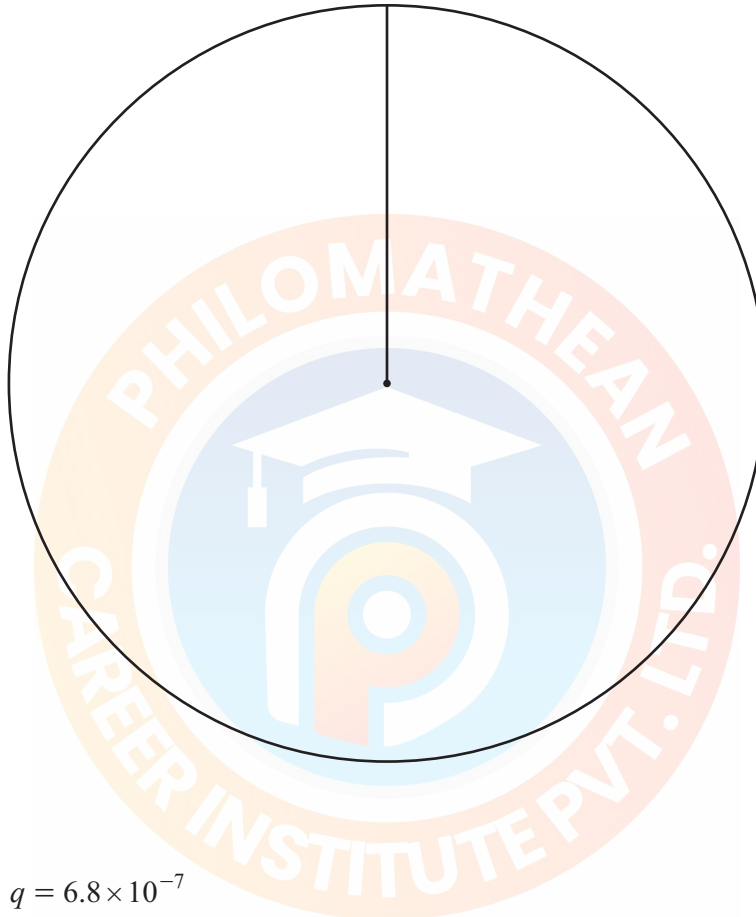
(b) $7(x - 3) - 2(2x - 1)$

..... [2]

- 5 One week a shop sells 120 jars of ketchup.
The table shows the number of jars of each size sold.

Size	Small	Medium	Large
Number sold	20	70	30

Show this information in a pie chart.



[3]

6 $p = 5 \times 10^{-8}$ $q = 6.8 \times 10^{-7}$

Find, giving your answers in standard form,

(a) pq

..... [2]

(b) $p + q$.

..... [2]

- 7 a , b and c are prime numbers.

$$V = a^2 b^4 c^3$$

$$W = a^5 b^3 c$$

Find the lowest common multiple (LCM) of V and W in terms of a , b and c .

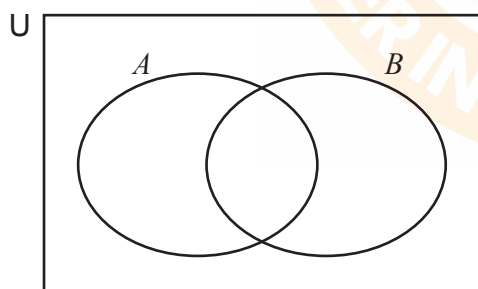
..... [2]

- 8 Jasmine drives a distance of a km at a speed of 50 km/h.
She then drives a distance of b km at a speed of 60 km/h.
She takes a total time of T hours for this journey.

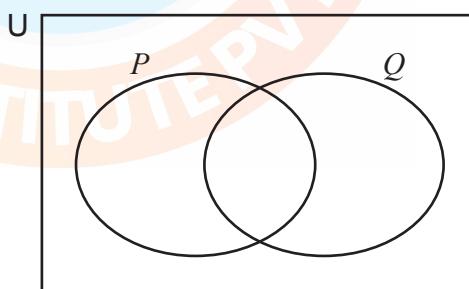
Find a formula for T in terms of a and b .

$T =$ [2]

- 9 In each Venn diagram, shade the given set.



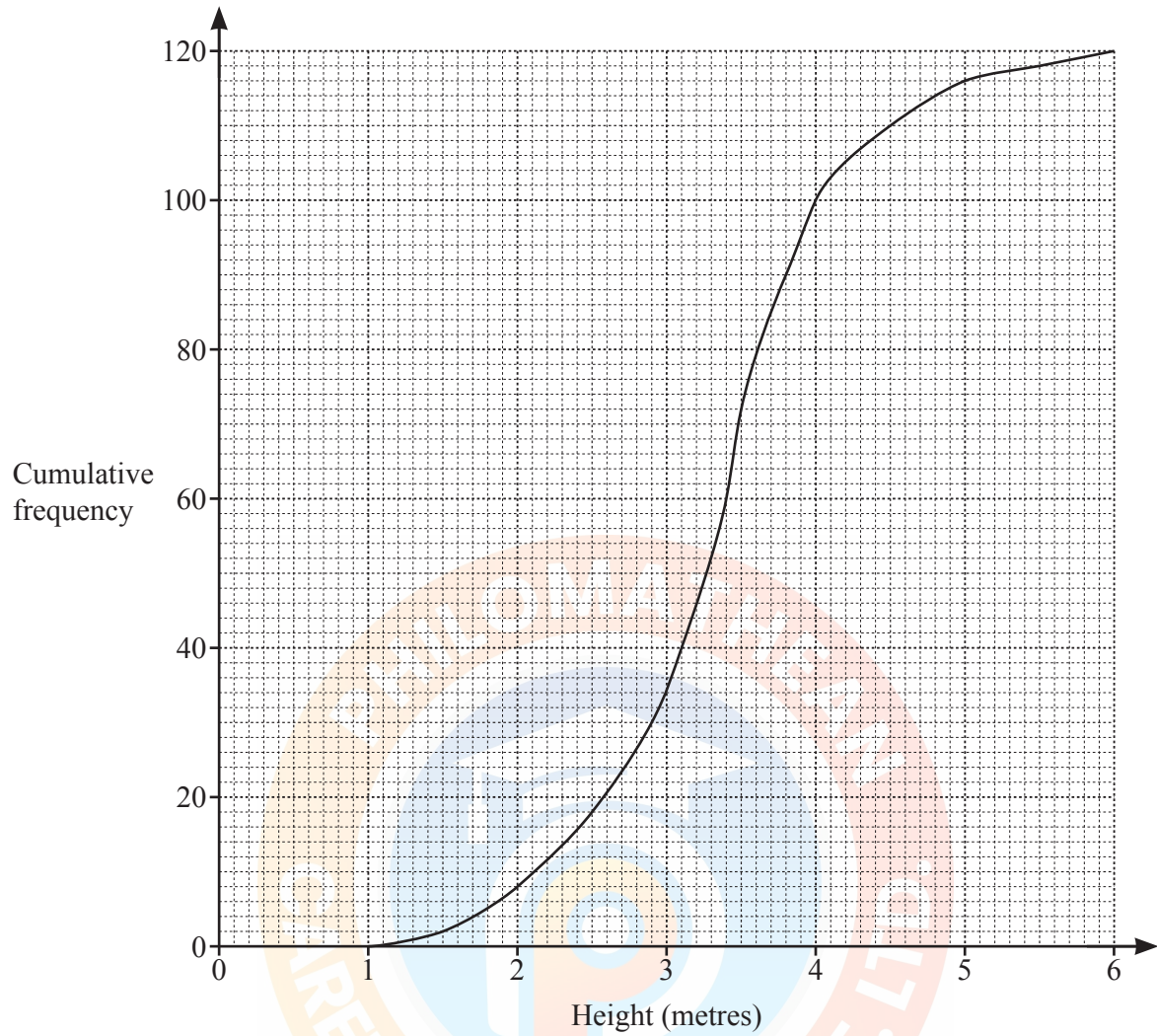
$A \cup B$



$(P \cap Q)'$

[2]

10 The cumulative frequency curve shows the heights of 120 young trees.



Find

(a) the median

..... m [1]

(b) the interquartile range.

..... m [2]

- 11 Factorise completely.

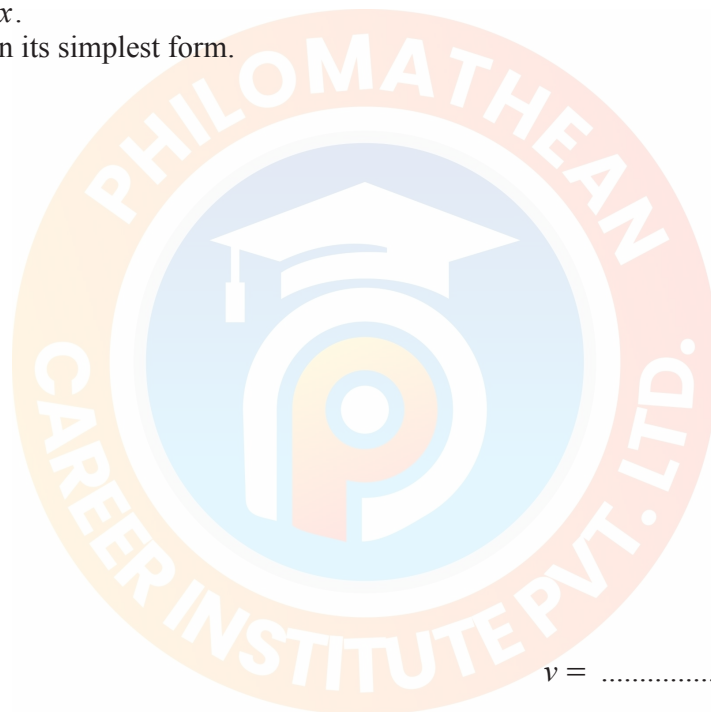
$$48x^2 - 75y^2$$

..... [3]

- 12 y is inversely proportional to the square root of x .
 v is directly proportional to y^2 .
 When $x = 9$, $y = 2$ and $v = 12$.

Find v in terms of x .

Give your answer in its simplest form.



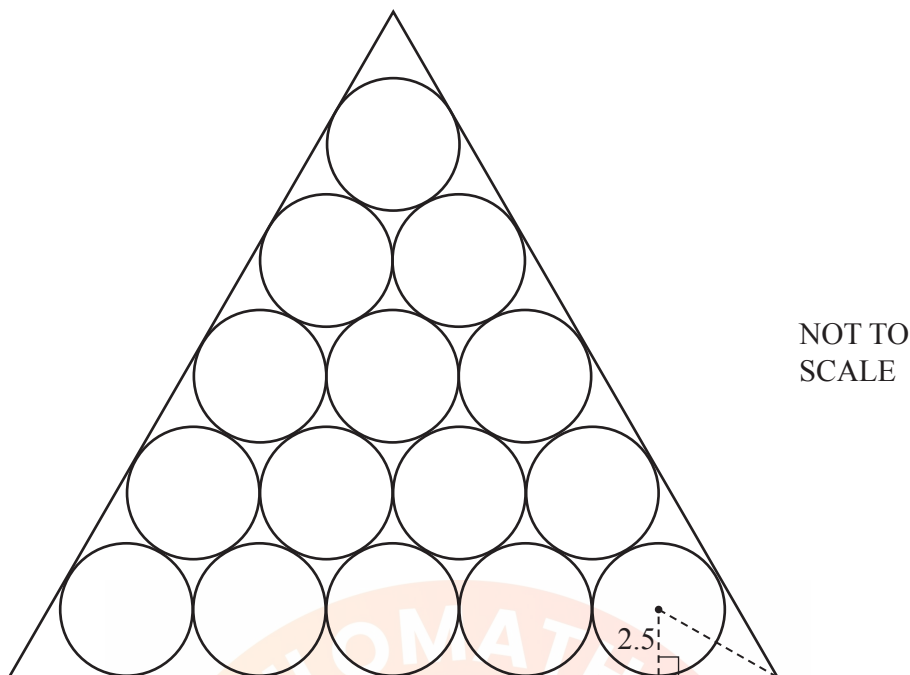
$v =$ [4]

- 13 Find the value of $4 \log 2 + 2 \log 5 - \log 4$.

..... [3]

Question 14 is printed on the next page.

14



The diagram shows 15 circles in an equilateral triangle.

The circles touch each other and the triangle.

The radius of each circle is 2.5 cm.

The length of each side of the triangle is $(a + b\sqrt{3})$ cm where a and b are integers.

Find the value of a and the value of b .

$a =$

$b =$ [5]

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0607/02

October/November 2012

45 minutes

Additional Materials: Geometrical Instruments

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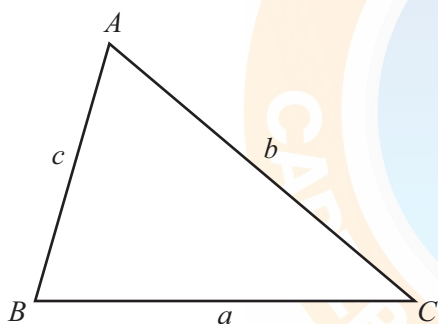
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$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

For
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Use

- 1 Factorise completely.

$$3xy - 6yz$$

Answer [2]

- 2 (a) Write 250 grams as a percentage of 2 kilograms.

Answer(a) % [2]

- (b) Manuel scores 46 in a test.
This is 15% more than his previous test score.

Calculate Manuel's previous test score.

Answer(b) [3]

- 3 Dariella leaves home at 07 49 and takes 24 minutes to walk to school.

- (a) At what time does Dariella arrive at school?

Answer(a) [1]

- (b) The distance to school is 1.4 km.

Calculate Dariella's walking speed.
Give your answer in kilometres per hour.

Answer(b) km/h [2]

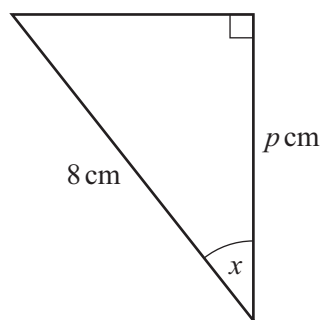
- 4 Calculate.

$$(3.24 \times 10^{-3}) \div (4 \times 10^4)$$

Give your answer in standard form.

Answer [2]

5 (a)

NOT TO
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$$\sin x = \frac{1}{3}$$

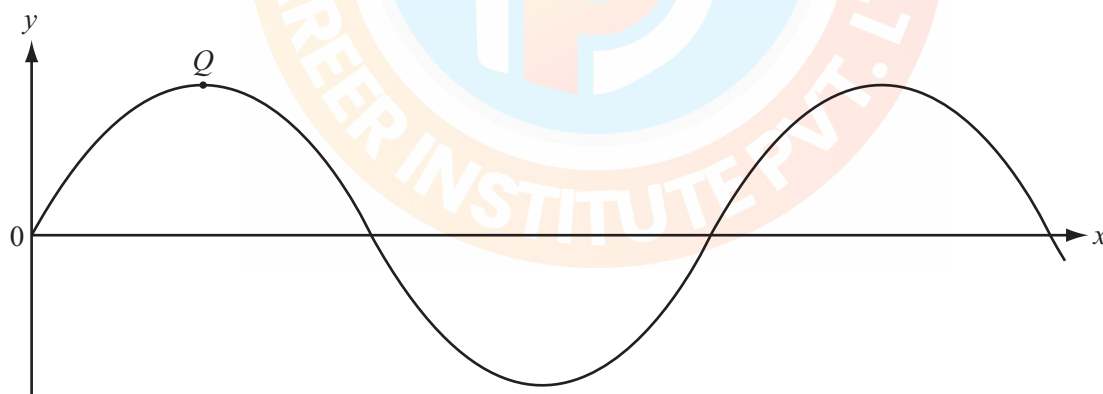
$$\cos x = \frac{2\sqrt{2}}{3}$$

$$\tan x = \frac{1}{2\sqrt{2}}$$

Calculate the value of p giving your answer as a simplified fraction.

Answer(a) $p =$ [2]

(b)



The diagram shows the graph of $y = 3 \sin 2x$.
 Q is a local maximum point.

Find the co-ordinates of Q .

Answer(b) (..... ,) [2]

6 (a) Simplify $\left(\frac{3}{2}\right)^{-3}$.

Give your answer as a fraction.

Answer(a) [2]

(b) $3 \log 2 - 2 \log 4 = \log t$

Find the value of t .

Answer(b) [2]

7 y varies inversely as the square root of x .
When $x = 16$, $y = 3$.

(a) Find y in terms of x .

Answer(a) $y =$ [2]

(b) Find y when $x = 36$.

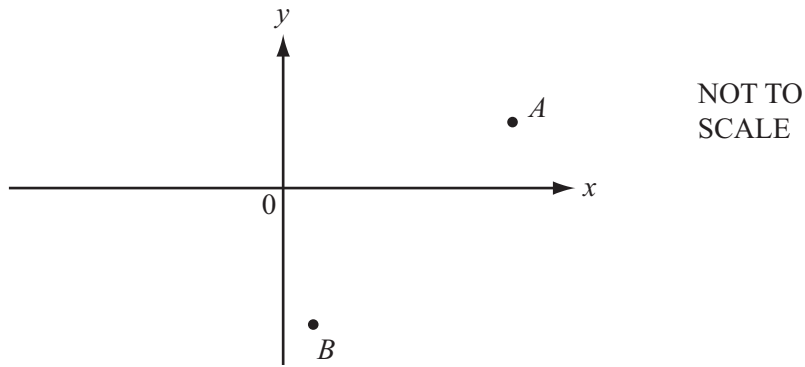
Answer(b) [1]

8 Write $1 - \frac{1}{x-1}$ as a single fraction.

Answer [2]

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9 (a)

For
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A is the point $(4, 2)$ and B is the point $(1, -3)$.

(i) Write down the vector $\vec{BA}$ in component form.

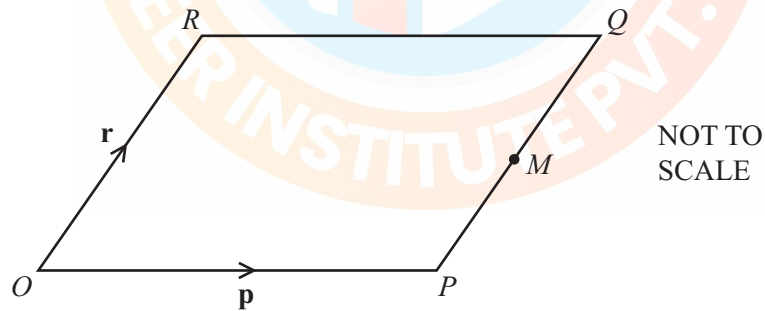
Answer(a)(i) $\vec{BA} = \begin{pmatrix} \\ \end{pmatrix}$ [1]

(ii) $\vec{BC} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$

Write down the co-ordinates of C .

Answer(a)(ii) $(\dots\dots\dots , \dots\dots\dots)$ [1]

(b)



$OPQR$ is a parallelogram and M is the midpoint of PQ .
 $\vec{OP} = \mathbf{p}$ and $\vec{OR} = \mathbf{r}$.

Find $\vec{OM}$ in terms of $\mathbf{p}$ and $\mathbf{r}$.

Answer(b) $\dots\dots\dots$ [2]

10 Simplify the following.

(a) $\sqrt{32}$

Answer(a) [1]

(b) $\frac{1}{\sqrt{2}+1}$

Answer(b) [2]

11



The diagrams show two similar shapes.
The lengths shown in the diagrams are in the ratio 2 : 1.

(a) Calculate the value of w .

Answer(a) $w =$ [1]

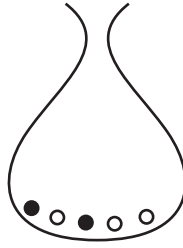
(b) The area of the larger shape is 56 cm^2 .

Calculate the area of the smaller shape.

Answer(b) cm^2 [2]

Question 12 is on the next page

12 (a)

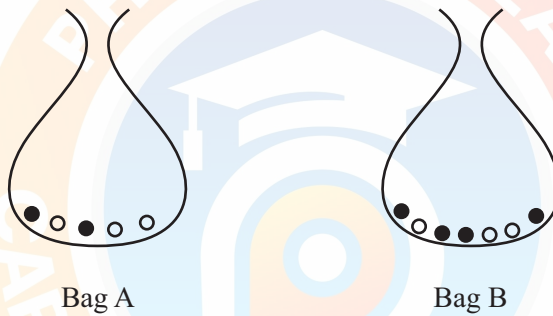


A bag contains 3 white beads and 2 black beads.
Two beads are taken out of the bag at random, without replacement.

Calculate the probability that both beads are white.

Answer(a) [2]

(b)



Bag A contains 3 white beads and 2 black beads.
Bag B contains 3 white beads and 4 black beads.

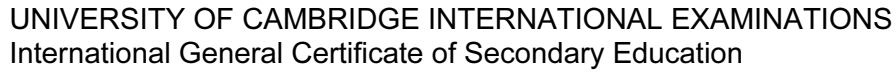
One bead is taken out of **each** bag at random.

Calculate the probability that one bead is white and one bead is black.

Answer(b) [3]

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0607/02

October/November 2013

45 minutes

Additional Materials: Geometrical Instruments

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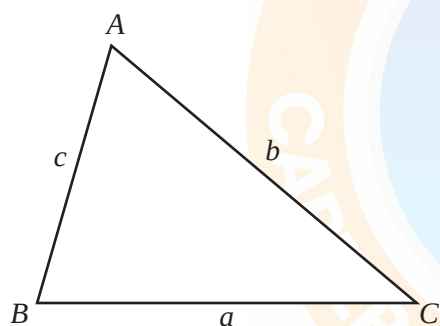
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Answer **all** the questions.

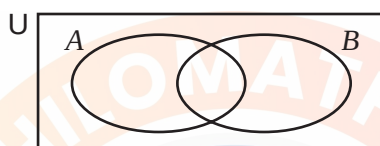
- 1 Solve the simultaneous equations.

$$\begin{aligned} 3g - 2h &= 11 \\ g - 2h &= 5 \end{aligned}$$

Answer $g =$

$h =$ [2]

- 2 (a)

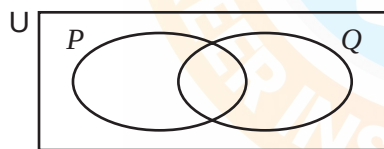


$$n(U) = 20, n(A \cup B)' = 3, n(A) = 11, n(B) = 13.$$

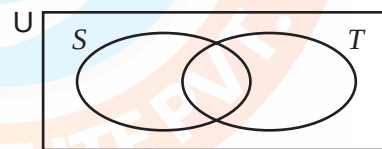
Find $n(A \cap B')$.

Answer(a) [2]

- (b) On each Venn diagram, shade the region indicated.



$(P \cap Q)'$



$S \cup T'$

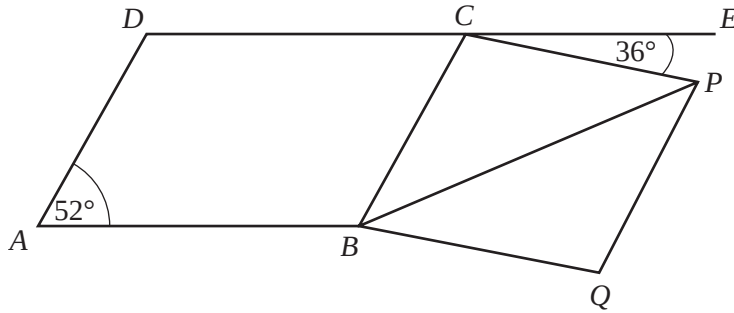
[2]

- 3 Tiago buys a concert ticket and then sells it for \$15.
He makes a profit of 20%.

Calculate how much Tiago paid for the ticket.

Answer \$ [3]

4

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$ABCD$ is a parallelogram and $BQPC$ is a rhombus.

DCE is a straight line.

Angle $DAB = 52^\circ$ and angle $ECP = 36^\circ$.

Find the size of angle BPC .

Answer [3]

5 (a) Simplify $\sqrt{72}$.

Answer(a) [1]

(b) $\frac{\sqrt{2}+2}{\sqrt{2}-1} = p + q\sqrt{2}$

Find the values of p and q .

Answer(b) $p =$

$q =$ [3]

6 Simplify the following.

(a) $2y^2 \times 3y^3$

Answer(a) [2]

(b) $\sqrt[3]{27p^{27}}$

Answer(b) [2]

7 (a) Find the amplitude and period of the function $f(x) = 4\cos(4x)$.

Answer(a) Amplitude =
Period = [2]

(b) $g(x) = 4\cos(4x) - 4$

Describe fully the **single** transformation that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

Answer(b)
..... [2]

- 8 (a) Write down the value of $8^{\frac{1}{3}}$.

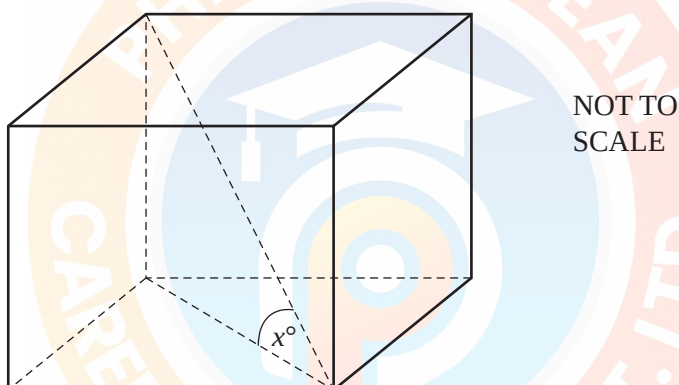
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Answer(a) [1]

- (b) Find the exact value of $\left(\frac{4}{3}\right)^{-2}$.

Answer(b) [2]

9

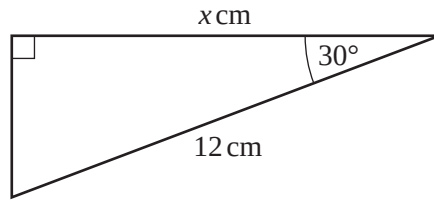


The diagram shows a cube of side length 1.

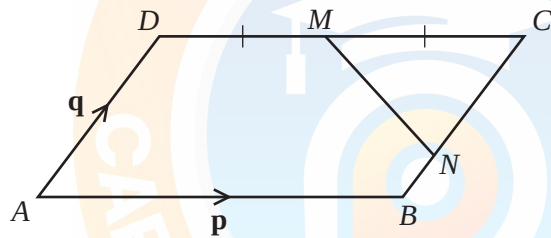
Find the value of $\tan x^\circ$.

Answer [3]

10

NOT TO
SCALEFind the exact value of x .For
Examiner's
UseAnswer $x =$ [3]

11

NOT TO
SCALE

$ABCD$ is a parallelogram.
 $DM = MC$ and $CN = 2NB$.
 $\vec{AB} = \mathbf{p}$ and $\vec{AD} = \mathbf{q}$.

(a) Write down $\vec{CN}$ in terms of $\mathbf{q}$.

Answer(a) [1]

(b) Find $\vec{MN}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

Answer(b) [1]

Question 12 is printed on the next page.

12 $f(x) = 3x - 1$

$g(x) = 12 - x$

Find

(a) $f(g(8))$,

Answer(a) [2]

(b) $f(g(x))$, in its simplest form,

Answer(b) [2]

(c) $g^{-1}(x)$.

Answer(c) $g^{-1}(x) =$ [1]For
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